

Polylogarithms: Global Euler Extremizers, Polynomial Zero Thresholds, and Mixed Gaussian Identities

A research continuation for the ProveIt project

10 October 2026

Abstract

This article continues the polylogarithm programme at the immutable ProveIt revision `4c173c06cc32c9cea554b39be1837ad2ae897fc1`, including the six archives present in its incoming directory. The emphasis is on precise outstanding extremal and zero-separation questions, with complete ordinary proofs and reproducible finite calculations. We prove the global scaling conjecture for the positive Euler remainder kernel, describe its eventual unique maximizer by a convergent expansion, and give an exact degree-seven characterization at truncation two. For the optimal order-averaged remainder we prove eventual exact reduction to the outer-order axis, settle the conjectured leading power law, and find a second scale of order $N^{-1.9695904\dots}/\sqrt{\log N}$. A polynomial lower bound $\text{mesh}(Q_n) \geq 1/(10n^2)$ for reciprocal-gamma Appell roots yields a uniform Lerch zero threshold of order $n^4 \log^2 n$ and a growing-index range of order $k^{1/4}/\sqrt{\log k}$. We classify the complete Gaussian sign transitions at outer orders -2 and -3 , prove cyclotomic sampling and elementary resummation identities for the mixed Gaussian generator, and propose a weight-fifteen relation with an exact rational residual bound below 10^{-775} . The last equality remains conjectural. An audit, integration notes, and fourteen research questions specify the remaining work.

Source snapshot. ProveIt revision `4c173c06cc32c9cea554b39be1837ad2ae897fc1`, including all six incoming archives present on 10 October 2026. Complete TeX sources, exact certificates, numerical diagnostics, and integration notes accompany this article.

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1 Scope, conventions, and the questions being continued

The canonical manuscript at the stated revision is a 379-page synthesis; its six incoming archives contain further overlapping continuations [1, 2, 3, 4, 5, 6, 7]. In particular, the S_4 Gaussian identity, the three recorded golden-ratio weight-five formulas, and the sharp Euler constant uniform in all positive orders have already been proved. They are background here. The existing S_6 and higher mixed Gaussian evaluations must still be assessed within their stated relation systems.

Our first point of departure is the positive remainder kernel

$$f_N(v) = \frac{(1-v)^N}{1+v}, \quad K_N(p, y) = \frac{f_N(py^2) - f_N(p)}{1-y}, \quad 0 \leq p, y \leq 1. \quad (1.1)$$

The quotient uses its continuous value at $y = 1$. The archive [2, Section 8] conjectures convergence of its global maxima to an exponential limiting kernel. A local scaling limit had been proved there; the missing global compactness argument and uniqueness of the limit are supplied below. Its finite-index uniqueness and monotonicity conjecture receives an exact answer at $N = 2$ and an eventual answer for large N .

The second point of departure is the zero geometry of the Stieltjes–Lerch derivative family. Its Laplace representation uses the Appell polynomials

$$\frac{e^{xz}}{\Gamma(1+z)} = \sum_{n \geq 0} Q_n(x) \frac{z^n}{n!}.$$

The effective incoming theorem [4, Section 4] uses an exponentially small lower bound on their root separation. Replacing that bound by a polynomial estimate changes the permitted growth of the Laurent index substantially.

Throughout, a multiple polylogarithm has decreasing summation indices:

$$\text{Li}_{a,b}(z, w) = \sum_{n > m \geq 1} \frac{z^n w^m}{n^a m^b}$$

in its domain of absolute convergence, followed by the specified analytic continuation. Put

$$F_{a,b}(z) = \text{Li}_{a,b}(z, 1) = \sum_{n \geq 1} \frac{H_{n-1}^{(b)}}{n^a} z^n, \quad H_m^{(b)} = \sum_{j=1}^m j^{-b}, \quad g_{a,b} = \text{Im } F_{a,b}(i). \quad (1.2)$$

For $a \geq 0, b > 0$ we use the principal continuation on $\mathbb{C} \setminus [1, \infty)$. Negative integer outer orders are obtained by powers of $\mathcal{D} = z d/dz$. The Dirichlet functions are

$$\beta(s) = \sum_{m \geq 0} \frac{(-1)^m}{(2m+1)^s}, \quad \eta(s) = \sum_{m \geq 1} \frac{(-1)^{m-1}}{m^s}$$

where these series converge, and their analytic continuations elsewhere. Our logarithms of positive real quantities are real logarithms.

Status of the mathematics. All statements marked theorem, proposition, lemma, or corollary are proved in this article, with any inherited input explicitly identified and justified as needed. The higher-weight relation marked conjecture has a frozen coefficient vector and an exact rational residual enclosure; an enclosure containing zero is not a proof of equality. None of the ordinary proofs or finite replay scripts is a proof-assistant formalization. The claims of advancement refer to the audited repository snapshot, not to an exhaustive assertion of priority over the literature.

2 Global extrema of the Euler remainder kernel

2.1 Why this kernel occurs

For $a, b > 0$, let S_a, T_b be independent gamma variables of shapes a, b and rate one, and put $X = e^{-S_a}$, $Y = e^{-T_b}$. At $a = 0$, take $X = 1$. Their moments are

$$\mathbb{E}X^{n-1} = n^{-a}, \quad \mathbb{E}Y^{m-1} = m^{-b}.$$

Writing the inner harmonic number as a geometric sum proves

$$F_{a,b}(z) = z^2 \mathbb{E} \frac{X}{(1 - zX)(1 - zXY)}. \quad (2.1)$$

Initially this follows by absolute convergence in the disk. On compact subsets of the principal slit plane, both denominators have positive uniform lower bounds, so the expectation is holomorphic and proves continuation. In particular,

$$g_{a,b} = -\mathbb{E} \frac{X^2(1 + Y)}{(1 + X^2)(1 + X^2Y^2)} < 0.$$

Define the finite Euler approximants by

$$A_j = \frac{H_{2j}^{(b)}}{(2j+1)^a}, \quad d_j = \sum_{m=0}^j (-1)^m \binom{j}{m} A_m, \quad E_N = \sum_{j=0}^{N-1} \frac{d_j}{2^{j+1}}. \quad (2.2)$$

The classical transformation and forward-difference convention are recorded in [8, Section 3.9(ii)]. Here the finite geometric identity gives an exact analytic remainder, even when the original boundary series is not convergent:

$$R_N(a, b) := 2^N (E_N - g_{a,b}) = \mathbb{E} K_N(X^2, Y) > 0. \quad (2.3)$$

Indeed,

$$d_j = \mathbb{E} \frac{(1 - X^2)^j - (1 - X^2Y^2)^j}{1 - Y},$$

and summing the finite geometric series proves (2.3). These steps retain the cancellation at $Y = 1$; they do not separate two divergent integrals.

The existing sharp bound uniform in all a, b, N is inherited from [1, 2, 3]. Our present problem is the different, pointwise optimization

$$M_N = \max_{(p,y) \in [0,1]^2} K_N(p, y). \quad (2.4)$$

An arbitrary maximizing point (p, y) need not be realized by the gamma family in (2.3). Thus M_N and the order-averaged constant studied in the next section are distinct extremal problems.

2.2 A uniform envelope and its limit

For $m \geq 2$, put

$$J_m(p, y) = \frac{(1 - py^2)^m - (1 - p)^m}{1 - y}.$$

Differentiation in p shows that its unique interior maximum for $0 < y < 1$ is attained at

$$p_m(y) = \frac{1 - y^{2/(m-1)}}{1 - y^{2m/(m-1)}}.$$

Substitution gives the envelope

$$J_m(p, y) \leq B_m(y) := (1 + y) \left(\frac{1 - y^2}{1 - y^{2m/(m-1)}} \right)^{m-1}. \quad (2.5)$$

The boundary values are $B_m(0) = 1$ and $B_m(1) = 2((m-1)/m)^{m-1}$. Moreover,

$$K_N(p, y) = \sum_{j \geq 0} 2^{-j-1} J_{N+j}(p, y). \quad (2.6)$$

This follows from the geometric expansion of $(1+v)^{-1}$ in powers of $1-v$. At $y=1$ it follows by differentiating that uniformly convergent geometric expansion, or by continuity.

The envelope calculation and its monotonicity were proved in [2, Section 3]. The following uniform quantitative version supplies the global limiting argument.

Lemma 2.1 (Uniform envelope convergence). *Define*

$$B(y) = (1 + y)y^{2y^2/(1-y^2)} \quad (0 < y < 1), \quad B(0) = 1, \quad B(1) = 2/e. \quad (2.7)$$

For every $m \geq 2$ and $0 \leq y \leq 1$,

$$B(y) \leq B_m(y) \leq B(y) \exp\left(\frac{1}{2(m-1)}\right). \quad (2.8)$$

For $0 < y < 1$, $B_{m+1}(y) < B_m(y)$. Consequently $K_N(p, y) \leq B_N(y)$ for all $N \geq 2$.

Proof. Set $t = -2 \log y > 0$, $k = m - 1$, and $h(v) = \log(1 - e^{-(1+v)t})$. Then

$$\log B_m(y) = \log(1 + y) - k\{h(1/k) - h(0)\}, \quad \log B(y) = \log(1 + y) - h'(0).$$

Here

$$h''(v) = -\left(\frac{t}{2 \sinh((1+v)t/2)}\right)^2, \quad -1 \leq h''(v) < 0.$$

Taylor's integral remainder therefore puts the difference of the two logarithms between zero and $1/(2k)$. Concavity also shows that the negative secant slope decreases as k increases, giving strict monotonicity. Continuity proves the endpoint inequalities. Finally average (2.5) in (2.6). $\square$

Lemma 2.2 (The unique limiting maximum). *The function B has a unique global maximizer y_∞ in $(0, 1)$. It is the unique root there of*

$$Q(y) = 1 + y - y^2 - y^3 + 4y \log y = 0. \quad (2.9)$$

The maximum is nondegenerate. Put

$$c_\infty = \frac{-2 \log y_\infty}{1 - y_\infty^2} = \frac{1 + y_\infty}{2y_\infty}, \quad M_\infty = (1 + y_\infty) \exp\left(-\frac{y_\infty(1 + y_\infty)}{2}\right). \quad (2.10)$$

Then $M_\infty > 1$, and

$$H(c, y) = \frac{e^{-cy^2} - e^{-c}}{1 - y}, \quad H(c, 1) = 2ce^{-c},$$

has the unique global maximizer (c_∞, y_∞) on $[0, \infty) \times [0, 1]$. Its Hessian there is negative definite.

Proof. Differentiating $\log B$ gives

$$(\log B)' = \frac{Q(y)}{(1-y^2)^2}, \quad Q''(y) = -6y - 2 + \frac{4}{y}.$$

The second derivative is positive on $(0, 2/3)$ and negative on $(2/3, 1)$. Since $Q'(0+) = -\infty$ and $Q'(1) = 0$, the derivative Q' has exactly one zero in $(0, 2/3)$; it is negative before that zero and positive after it, up to one. Also $Q(0+) = 1$ and $Q(1) = 0$. Thus Q crosses zero exactly once inside $(0, 1)$, from positive to negative, and $Q'(y_\infty) < 0$. This proves uniqueness and nondegeneracy of the maximum. Its value exceeds one, for example because $B(1/4) > 1$ is equivalent to the integer inequality $5^{15} > 2^{34}$.

For fixed $0 < y < 1$, differentiating H in c gives its unique maximum at $c = -2 \log y / (1 - y^2)$, with value $B(y)$. At $y = 0$ its supremum is one, at $y = 1$ its maximum is $2/e$, and at $c = 0$ it is zero. This proves the two-variable uniqueness. Equation (2.9) gives the second expression for c_∞ in (2.10). At a stationary point in c ,

$$H_{cc} = -y^2(1+y)e^{-cy^2} < 0.$$

The Schur complement of this entry in the Hessian is the second derivative of the optimized profile B , which is strictly negative at y_∞ . Hence the Hessian is negative definite. $\square$

2.3 Resolution of the global limit conjecture

Theorem 2.3 (Global limiting extrema). *For every $N \geq 2$,*

$$M_\infty < M_N \leq M_\infty \exp\left(\frac{1}{2(N-1)}\right). \quad (2.11)$$

In particular, $M_N \rightarrow M_\infty$. If (p_N, y_N) is any global maximizer of K_N , then

$$Np_N \rightarrow c_\infty, \quad y_N \rightarrow y_\infty. \quad (2.12)$$

Thus the global limit conjecture of [2, Section 8, conj:limit] holds, including its assertion about all maximizing points.

Proof. For the strict lower bound set $A_N(v) = -\log f_N(v) = -N \log(1-v) + \log(1+v)$ on $[0, 1)$. It is strictly convex on $(0, 1)$ because

$$A_N''(v) = \frac{N}{(1-v)^2} - \frac{1}{(1+v)^2} > 0.$$

There is a unique $p \in (0, 1)$ such that $A_N(p) = c_\infty$. Strict convexity and $A_N(0) = 0$ imply $A_N(py_\infty^2) < y_\infty^2 A_N(p)$, and therefore

$$K_N(p, y_\infty) > H(c_\infty, y_\infty) = M_\infty.$$

The upper bound follows from lemma 2.1.

Uniform envelope convergence and uniqueness of the maximum of B force $y_N \rightarrow y_\infty$: any closed subset avoiding that point has an envelope maximum bounded strictly below M_∞ for large N . Write $c_N = Np_N$. Since y_N stays away from zero and one,

$$K_N(p_N, y_N) \leq \frac{(1 - p_N y_N^2)^N}{1 - y_N} \leq \frac{e^{-c_N y_N^2}}{1 - y_N}$$

rules out $c_N \rightarrow \infty$, and indeed bounds all c_N . For bounded c , the functions $f_N(c/N)$ and their derivatives in c converge uniformly to e^{-c} . The divided-difference integral

$$K_N(c/N, y) = -c(1+y) \int_0^1 \frac{d}{dv} f_N(v/N) \Big|_{v=cy^2+uc(1-y^2)} du$$

proves uniform convergence to H on compact rectangles, including $y = 1$. Every subsequential limit of (c_N, y_N) must consequently maximize H . Its unique maximizing point gives (2.12). $\square$

Remark 2.4 (What supplies global compactness). The local scaling limit alone does not rule out maximizing points with $Np \rightarrow \infty$ or $y \rightarrow 0$. The uniform envelope first localizes y ; only then does the exponential estimate localize Np . This is the additional argument required by the incoming conjecture.

2.4 A convergent expansion and eventual strict monotonicity

Set $x = (c, y)$ and $x_0 = (c_\infty, y_\infty)$. Introduce

$$\mathcal{F}(\varepsilon, v) = \exp \left(\frac{\log(1 - \varepsilon v)}{\varepsilon} - \log(1 + \varepsilon v) \right), \quad (2.13)$$

$$\mathcal{K}(\varepsilon, c, y) = \frac{\mathcal{F}(\varepsilon, cy^2) - \mathcal{F}(\varepsilon, c)}{1 - y}, \quad (2.14)$$

with the removable value $\mathcal{F}(0, v) = e^{-v}$. These functions are real analytic in a neighborhood of $(0, x_0)$ and have holomorphic extensions to a complex neighborhood there.

Theorem 2.5 (Convergent extremizer expansion). *For all sufficiently large integers N , K_N has a unique global maximizer in $(0, 1)^2$. There are convergent power series near $\varepsilon = 0$,*

$$x(\varepsilon) = x_0 + \sum_{j \geq 1} x_j \varepsilon^j, \quad M(\varepsilon) = M_\infty + \sum_{j \geq 1} \mu_j \varepsilon^j, \quad (2.15)$$

such that

$$(Np_N, y_N) = x(1/N), \quad M_N = M(1/N).$$

The first coefficient is

$$\mu_1 = \frac{1}{2} M_\infty c_\infty^2 y_\infty^2 > 0. \quad (2.16)$$

Consequently $M_{N+1} < M_N$ for all sufficiently large N .

Proof. The gradient of $\mathcal{K}$ in x vanishes at $(0, x_0)$, and its Jacobian in x is the invertible Hessian in lemma 2.2. The analytic implicit-function theorem gives the convergent series $x(\varepsilon)$. On a sufficiently small convex neighborhood of x_0 , the Hessian remains negative definite for small ε , so there is at most one stationary maximum there. By theorem 2.3, every global maximizer lies in that neighborhood for large N . Thus the analytic stationary point is the unique global maximizer. Composition gives $M(\varepsilon)$.

The first-order coefficient of $\mathcal{F}(\varepsilon, v)$ is $-e^{-v}(v + v^2/2)$. Since $\nabla H(x_0) = 0$, the coefficient μ_1 is obtained by evaluating the first-order coefficient of $\mathcal{K}$ at x_0 . The relation $e^{-c_\infty} = y_\infty^2 e^{-c_\infty y_\infty^2}$ cancels its linear terms and gives (2.16). Analyticity then implies $M'(\varepsilon) > 0$ for sufficiently small positive ε . Replacing $\varepsilon = 1/N$ by the smaller value $1/(N+1)$ proves the assertion. $\square$

All coefficients have an explicit finite recursion. Define

$$A_j(v) = -\frac{v^{j+1}}{j+1} + \frac{(-1)^j v^j}{j}, \quad P_0(v) = 1, \quad P_j(v) = \frac{1}{j} \sum_{\ell=1}^j \ell A_\ell(v) P_{j-\ell}(v). \quad (2.17)$$

Then

$$\mathcal{F}(\varepsilon, v) = e^{-v} \sum_{j \geq 0} P_j(v) \varepsilon^j, \quad P_1 = -v - v^2/2, \quad P_2 = v^2 + v^3/6 + v^4/8.$$

Put

$$H_j(c, y) = \frac{e^{-cy^2} P_j(cy^2) - e^{-c} P_j(c)}{1 - y}, \quad J = \nabla^2 H_0(x_0), \quad q = \nabla H_1(x_0).$$

In particular,

$$x_1 = -J^{-1}q, \quad \mu_2 = H_2(x_0) - \frac{1}{2}q^\top J^{-1}q. \quad (2.18)$$

At every subsequent order, equating coefficients in $\nabla_x \mathcal{K}(\varepsilon, x(\varepsilon)) = 0$ gives the same invertible linear system $Jx_j =$ an already known vector.

The accompanying calculations give the following diagnostic decimals:

$$\begin{aligned} y_\infty &= 0.1453038592593584238735558631 \dots, \\ c_\infty &= 3.9410648316472506681136817940 \dots, \\ M_\infty &= 1.0538619303627116260728728105 \dots, \\ \mu_1 &= 0.1727965939279004260875754084 \dots, \\ \mu_2 &= -0.3325546448598136087570212772 \dots, \\ (x_1)_c &= -12.6952916582438127590265516499 \dots, \\ (x_1)_y &= 0.3696596852920567080244169744 \dots \end{aligned} \quad (2.19)$$

The rational verification script separately encloses y_∞ , c_∞ , M_∞ , and μ_1 to 45 decimal places; high-precision derivative evaluations of the later constants are diagnostics. The formulas (2.16) and (2.18), not their decimals, define the coefficients.

Remark 2.6 (Remaining finite-index question). The theorem proves eventual uniqueness and monotonicity without giving an explicit first valid integer. It does not settle the incoming claim for every $N \geq 2$. The following exact base case is useful, but does not by itself close that finite-index gap.

2.5 An exact degree-seven extremum at truncation two

Lemma 2.7 (Interior location). *Every global maximizer of K_N lies in $(0, 1)^2$ when $N \geq 2$.*

Proof. The maximum exceeds $M_\infty > 1$. On $p = 0$ the kernel is zero, and on $y = 0$ it is $1 - f_N(p) \leq 1$. On $p = 1$, $0 < y < 1$,

$$\partial_p K_N(1, y) = \frac{y^2 f'_N(y^2)}{1 - y} < 0,$$

since $f'_N(1) = 0$ for $N \geq 2$. Thus moving inward improves the value. On $y = 1$ the value is $-2pf'_N(p)$, with zero values at both p -endpoints. At any interior maximum of this boundary function, $f'_N(p) + pf''_N(p) = 0$. Taylor expansion in y gives

$$\partial_y K_N(p, 1) = -pf'_N(p) - 2p^2 f''_N(p) = pf'_N(p) < 0.$$

Moving inward again improves the value. All boundary possibilities are thereby excluded. $\square$

Theorem 2.8 (Exact first interior maximum). *Let y_2 be the unique root in $(0, 1)$ of*

$$G(y) = y^7 + 3y^6 - 2y^5 - 10y^4 + 17y^3 + 59y^2 - 4. \quad (2.20)$$

Define

$$D(y) = y^6 + 2y^5 - 2y^4 - 6y^3 + 17y^2 + 36y + 4, \quad p_2 = \frac{12}{D(y_2)}. \quad (2.21)$$

Then (p_2, y_2) is the unique global maximizer of K_2 , and

$$M_2 = p_2(1 + y_2) \left(\frac{4}{(1 + p_2)(1 + p_2 y_2^2)} - 1 \right). \quad (2.22)$$

In particular, this is an exact algebraic specification, not a numerical stationary-point search.

Proof. Polynomial division gives

$$K_2(p, y) = p(1 + y) \left(\frac{4}{(1 + p)(1 + py^2)} - 1 \right).$$

After multiplying its gradient equations by positive denominators and removing the positive factors $1 + y$ and p , respectively, the equations are $P(p, y) = Q_2(p, y) = 0$, where

$$\begin{aligned} P &= -p^4 y^4 - 2p^3 y^4 - 2p^3 y^2 - p^2 y^4 - 8p^2 y^2 - p^2 - 2py^2 - 2p + 3, \\ Q_2 &= -p^3 y^4 - p^2 y^4 - 2p^2 y^2 - 6py^2 - 8py - p + 3. \end{aligned}$$

Their last two nonzero subresultants in the variable p are

$$-16y^{10}\{pD(y) - 12\}, \quad 192y^{10}(y + 1)G(y). \quad (2.23)$$

These identities follow by the polynomial Euclidean algorithm; the supplied exact replay reconstructs the polynomials from the kernel and checks both subresultants. A common zero in the open square must therefore have $G(y) = 0$ and $pD(y) = 12$.

On $(0, 1)$,

$$\frac{G'(y)}{y} = 7y^5 + 18y^4 - 10y^3 - 40y^2 + 51y + 118 > 68.$$

Also $G(0) = -4$ and $G(1) = 64$. Hence there is exactly one possible y -coordinate for an interior stationary point, and (2.21) gives exactly one possible p -coordinate. An interior global maximum exists by compactness and lemma 2.7. Thus that stationary point exists, is unique, and is the global maximum. No reverse implication from a potentially extraneous resultant root is needed. $\square$

Reproducible numerical comparisons. The exact rational replay encloses the algebraic solution and proves, in particular,

$$\begin{aligned} M_2 &\geq 1.118938685961052674777466790900108430060964540, \\ M_2 &\leq 1.118938685961052674777466790900108430060964541. \end{aligned}$$

Figure 1 plots the analogous values $N(\widetilde{M}_N - M_\infty)$ obtained by local stationary-point computation against $\mu_1 + \mu_2/N$. Its finite-index entries are explicitly marked as diagnostics; the global theorems above do not rely on the completeness of those numerical searches.

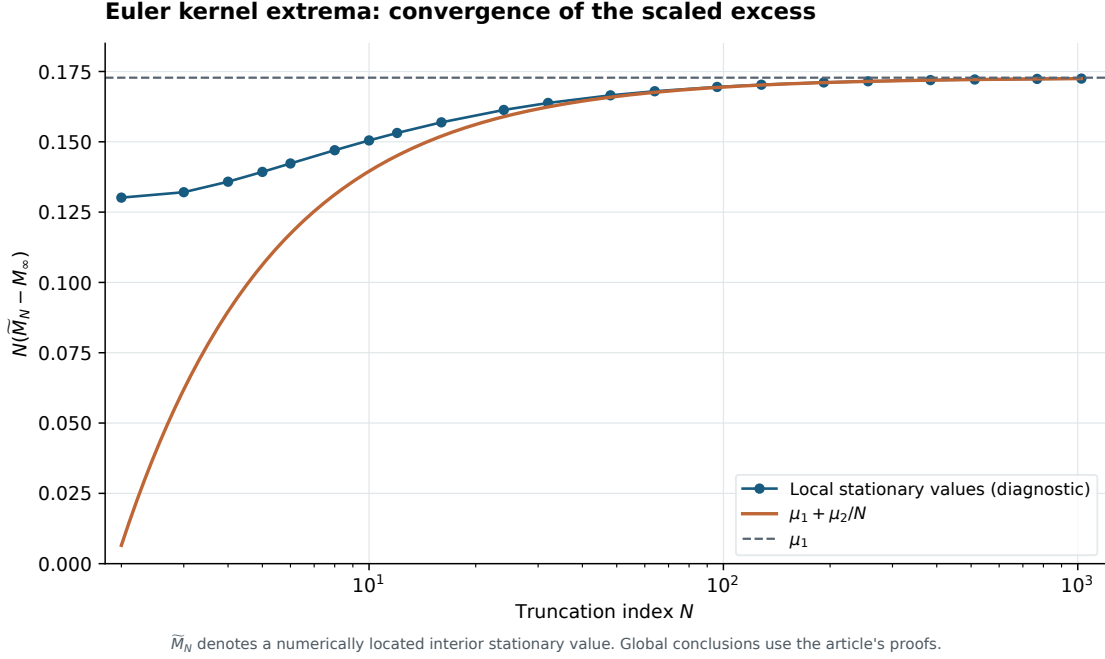


Figure 1: Approach to the first two correction terms. The plotted $\widetilde{M}_N$ are values at numerically found interior stationary points, with local Hessians checked. They are diagnostics, not additional global certificates at the plotted finite indices. The global large- N assertion and the coefficients are proved in Theorem 2.5.

3 The optimal order-averaged Euler constant

The incoming report [3, Sections 2 and 9] proves that the optimal order-uniform scaled remainder tends to one and gives a matching candidate lower asymptotic. Its upper asymptotic remains an explicit conjecture there. We prove that conjecture, identify the eventual maximizing boundary, and determine a further asymptotic scale.

Write

$$d_j(a, b) = \sum_{n=0}^j (-1)^n \binom{j}{n} \frac{H_{2n}^{(b)}}{(2n+1)^a}, \quad E_N(a, b) = \sum_{j=0}^{N-1} \frac{d_j(a, b)}{2^{j+1}},$$

$$R_N(a, b) = 2^N (E_N(a, b) - g_{a,b}).$$

Let T_a, U_b be independent gamma variables of rate one and shapes a, b , with $T_0 = 0$. To avoid confusing the logarithmic kernel with the polylogarithm $F_{a,b}$, put

$$\Psi_N(t) = f_N(e^{-2t}), \quad J_N(s) = \mathbb{E} \Psi_N(T_s), \quad J_N(0) = 0. \quad (3.1)$$

The existing positive two-measure identity is

$$R_N(a, b) = \mathbb{E} \frac{\Psi_N(T_a + U_b) - \Psi_N(T_a)}{1 - e^{-U_b}}. \quad (3.2)$$

It holds for $a \geq 0, b > 0$; in particular it does not require a finite unweighted signed moment measure. Expanding the denominator into a positive geometric series and using gamma addition gives, for $b > 1$,

$$0 \leq R_N(a, b) - \{J_N(a+b) - J_N(a)\} \leq \zeta(b) - 1. \quad (3.3)$$

For completeness, multiplication of the density of U_b by $e^{-(r-1)U_b}$ gives r^{-b} times a gamma density of rate r . The term $r = 1$ is the displayed difference of J_N values, while each remaining expectation is between zero and one. Tonelli's theorem justifies the sum.

We also use the previously proved bound

$$0 < R_N(a, b) < 1 \quad (a \geq 1, b > 0), \quad (3.4)$$

from the manuscript's extended one-sided Euler theorem [1, Chapter 5, 05-real-euler.tex]. We identify this input explicitly: the universal constant greater than one alone would not replace it in the proof below.

3.1 Eventual reduction to the outer-order axis

Define

$$A_N(b) = R_N(0, b) = \frac{1}{\Gamma(b)} \int_0^\infty t^{b-1} e^{-t} k_N(e^{-t}) dt, \quad k_N(v) = \frac{(1+v)(1-v^2)^{N-1}}{1+v^2}. \quad (3.5)$$

All limits of truncation indices in this section are through positive integers.

Theorem 3.1 (Eventual exact axis reduction). *For all sufficiently large N ,*

$$C_N := \sup_{a, b > 0} R_N(a, b) = \max_{b > 0} A_N(b) > 1. \quad (3.6)$$

The supremum over positive orders is not attained. If $a = 0$ is admitted, every global maximizer is on that axis. Moreover, $C_{N+1} < C_N$ for all sufficiently large N .

Proof. Put $L = \log N$. We separate $b \geq 3L/2$ from $b \leq 3L/2$, always assuming N large enough for the indicated intervals to be nonempty.

Comparison above the cutoff. The function f_N is decreasing and convex on $[0, 1]$, and $0 \leq -f'_N(x) \leq N + 1$. Thus, for $0 \leq r, v \leq 1$,

$$f_N(rv^2) - f_N(v^2) \leq (N + 1)(1 - r)v^2.$$

Subtract the axis integrand from (3.2) and set $r = e^{-2T_a}$, $v = e^{-U_b}$. Independence gives the useful uniform comparison

$$\begin{aligned} R_N(a, b) - A_N(b) &\leq (N + 1)(1 - 3^{-a})\{\zeta(b) - 1 - 2^{-b}\} \\ &\quad - J_N(a)\zeta(b), \quad b > 1. \end{aligned} \quad (3.7)$$

Indeed $\mathbb{E}(1 - r) = 1 - 3^{-a}$, $\mathbb{E}[v^2/(1 - v)] = \zeta(b) - 1 - 2^{-b}$, and $\mathbb{E}[1/(1 - v)] = \zeta(b)$.

For $0 < a \leq 1$ there is an elementary lower bound

$$J_N(a) \geq \frac{ae^{-2}}{L + 4} N^{-1/2}. \quad (3.8)$$

To prove it, restrict the defining integral to $t \in [L/2 + 1, L/2 + 2]$. Bernoulli's inequality gives

$$\Psi_N(t) \geq 1 - (N + 1)e^{-2t} \geq 1 - 2e^{-2} > \frac{1}{2}.$$

Log convexity of the gamma function gives $\Gamma(a + 1) \leq 1$ for $0 \leq a \leq 1$, and hence $1/\Gamma(a) \geq a$. Also $t^{a-1} \geq 1/t$ on the interval in question, and $e^{-t} \geq e^{-2}N^{-1/2}$. Its length is one, proving (3.8).

For $b \geq 2$, integral comparison gives

$$\zeta(b) - 1 - 2^{-b} \leq 3^{-b} \left(1 + \frac{3}{b-1}\right) \leq 4 \cdot 3^{-b}.$$

Consequently the positive term in (3.7), for $0 < a \leq 1$ and $b \geq 3L/2$, is at most

$$8a \log 3 N^{1-(3/2)\log 3}.$$

Its ratio to the right side of (3.8) is bounded by

$$8e^2 \log 3 (L+4) e^{-(3/2)(\log 3-1)L} \rightarrow 0.$$

We have therefore proved, uniformly on this parameter region,

$$R_N(a, b) \leq A_N(b) - \frac{ae^{-2}}{2(L+4)} N^{-1/2} \quad (0 < a \leq 1, b \geq 3L/2) \quad (3.9)$$

for all sufficiently large N .

Exclusion below the cutoff. Monotonicity of J_N in its gamma shape and (3.3) give

$$R_N(a, b) \leq J_N(b+1) + \zeta(b) - 1 \quad (0 \leq a \leq 1, b > 1). \quad (3.10)$$

Since $\Psi_N(L/2) = f_N(1/N) \leq e^{-1}$,

$$1 - J_N(s) \geq (1 - e^{-1}) \Pr\{T_s \leq L/2\}. \quad (3.11)$$

For every fixed $D > 1/2$ there is $c_D > 0$ such that

$$\Pr\{T_{DL+1} \leq L/2\} \geq c_D L^{-1/2} e^{-I(D)L}, \quad I(D) = \frac{1}{2} - D + D \log(2D), \quad (3.12)$$

for all sufficiently large L . This lower bound follows directly by integrating the gamma density over $[L/2 - 1, L/2]$:

$$\Pr\{T_{DL+1} \leq L/2\} \geq \frac{(L/2 - 1)^{DL} e^{-L/2}}{\Gamma(DL + 1)}.$$

Stirling's formula [8, Section 5.11] supplies (3.12); no large-deviation equivalence is needed.

On $4 \leq b \leq L/2$, gamma stochastic ordering and the central limit theorem imply that the lower bound in (3.11) eventually exceeds $1/5$ uniformly, since at the worst endpoint $\Pr\{T_{L/2+1} \leq L/2\} \rightarrow 1/2$. The central limit theorem here follows from

$$\mathbb{E} e^{it(T_s-s)/\sqrt{s}} = e^{-it\sqrt{s}} (1 - it/\sqrt{s})^{-s} \rightarrow e^{-t^2/2}.$$

On the other hand $\zeta(b) - 1 \leq \zeta(4) - 1 \leq 5/48 < 1/5$. On $L/2 \leq b \leq L$, apply (3.12) with $D = 1$. The deficit is bounded below by a positive constant times

$$L^{-1/2} N^{-(\log 2 - 1/2)},$$

whereas $\zeta(b) - 1 \leq 2N^{-(\log 2)/2}$. The former dominates because $\log 2 - 1/2 < (\log 2)/2$. On $L \leq b \leq 3L/2$, take $D = 3/2$. The corresponding deficit is at least a positive constant times

$$L^{-1/2} N^{-((3/2)\log 3 - 1)},$$

whereas $\zeta(b) - 1 \leq 2N^{-\log 2}$. Again the deficit dominates: $(3/2)\log 3 - 1 < \log 2$. The last inequality is equivalent to $3\sqrt{3} < 2e$; for example $e > 8/3$ and $3\sqrt{3} < 16/3$ prove it. Thus (3.10) is strictly below one throughout $4 \leq b \leq 3L/2$ for large N .

The bounded rectangle $0 \leq a \leq 1$, $0 < b \leq 4$ is also uniform. Couple $T_a \leq T_1$ and $U_b \leq U_4$. The derivative satisfies $0 \leq \Psi'_N \leq 4$, and its supremum on every bounded interval tends to zero. Indeed, with $v = e^{-2t}$, its two positive terms are bounded by $2Nv(1-v)^{N-1} \leq 2$ and

$2v/(1+v)^2 \leq 1/2$. The actual terms contain $(1-v)^{N-1}$ and $(1-v)^N$, respectively, which give uniform decay on every fixed bounded t -interval. Since $u/(1-e^{-u}) \leq 1+u$, we obtain

$$R_N(a, b) \leq \mathbb{E} \left[(1 + U_4) \sup_{0 \leq t \leq T_1 + U_4} \Psi'_N(t) \right].$$

The integrand tends to zero and is bounded by $4(1 + U_4)$. Dominated convergence therefore proves uniform convergence to zero on the bounded rectangle and completes the exclusion below $3L/2$.

Finally, (3.4) excludes $a \geq 1$. The function A_N is continuous, tends to zero as $b \downarrow 0$ for $N \geq 2$, and tends to one as $b \rightarrow \infty$. The elementary expansion established below shows that $A_N(b) > 1$ for all sufficiently large b at each fixed N . Its maximum is therefore attained and exceeds one. Every positive outer order is strictly inferior either to one or to its corresponding axis value, while continuity as $a \downarrow 0$ approaches each axis value. Finally, $A_{N+1}(b) < A_N(b)$ at every $b > 0$ by its positive integral. If b' maximizes A_{N+1} and both indices are in the proved range, then

$$C_{N+1} = A_{N+1}(b') < A_N(b') \leq C_N.$$

This proves eventual strict monotonicity as well. $\square$

3.2 Resolution of the asymptotic-constant conjecture

Set

$$\begin{aligned} h &= \log(3/2), & \alpha &= h^{-1}, & p &= \frac{\log 2}{h}, & q &= \frac{\log 3}{\log 2}, \\ d_0 &= \alpha \log q, & \hat{b}_N &= \alpha \log N + d_0 = \frac{\log(Nq)}{h}, & c &= (1 - q^{-1})q^{-p}. \end{aligned} \quad (3.13)$$

Lemma 3.2 (Axis sign changes and a Taylor enclosure). *For $N \geq 2$, $k_N(v) - 1$ has exactly one zero in $(0, 1)$, with positive sign before it and negative sign after it. Consequently, if $A_N(b_0) < 1$, then $A_N(b) < 1$ for $0 < b < b_0$. For every $N \geq 1$ and $b > 0$,*

$$\begin{aligned} A_N(b) &= 1 + 2^{-b} - N3^{-b} - N4^{-b} + \varepsilon_N(b), \\ 0 &\leq \varepsilon_N(b) \leq \frac{N^2 - N + 2}{2}(5^{-b} + 6^{-b}). \end{aligned} \quad (3.14)$$

Proof. The numerator determining the sign of the logarithmic derivative of k_N on $(0, 1)$ is

$$P_N(v) = 1 - (2N + 1)v + v^2 + (3 - 2N)v^3.$$

For $N \geq 2$, its derivative is strictly negative on $[0, 1]$. Since $P_N(0) > 0 > P_N(1)$, the function k_N first increases and then decreases. Its endpoint values are one and zero, proving the claimed single crossing. In the variable $t = -\log v$, the signed function $k_N(e^{-t}) - 1$ is negative before a unique $t_0 > 0$ and positive after it. If $b < b_0$, multiplication by t^{b-b_0} changes its integral by at most multiplication by $t_0^{b-b_0}$: the factor is larger on the negative part and smaller on the positive part. This proves the implication for A_N after normalizing by the positive gamma factors.

For the enclosure put $G_N(x) = (1-x)^{N-1}/(1+x)$. Its second derivative is nonnegative and bounded by $N^2 - N + 2$ on $[0, 1]$. For $N \geq 3$,

$$G_N''(x) = \frac{(N-1)(N-2)(1-x)^{N-3}}{1+x} + \frac{2(N-1)(1-x)^{N-2}}{(1+x)^2} + \frac{2(1-x)^{N-1}}{(1+x)^3};$$

$N = 1, 2$ are checked directly. Since $G_N(0) = 1$ and $G'_N(0) = -N$, Taylor's integral remainder gives

$$0 \leq G_N(x) - 1 + Nx \leq \frac{N^2 - N + 2}{2} x^2.$$

Multiply by $1 + v$, substitute $x = v^2$, and use $\mathbb{E}e^{-jU_b} = (j + 1)^{-b}$. $\square$

Theorem 3.3 (Sharp late Euler constant and unique axis maximizer). *The conjecture of [3, Section 9] holds:*

$$C_N - 1 \sim cN^{-p}. \quad (3.15)$$

For all sufficiently large N , A_N has a unique global maximizer b_N , and

$$b_N - \hat{b}_N \rightarrow 0. \quad (3.16)$$

Proof. The elementary function $2^{-b} - N3^{-b}$ has its unique maximum at $b = \hat{b}_N$, where its value is cN^{-p} . At that point the remaining terms in (3.14) are $o(N^{-p})$:

$$N4^{-\hat{b}_N} = O(N^{1-2p}), \quad N^2 5^{-\hat{b}_N} = O(N^{2-\log 5/h}).$$

Here $p > 1$, and $\log 5/h > p + 2$ is equivalent to $5 > 9/2$. This proves the required lower asymptotic.

For the upper bound, take $L = \log N$ and first evaluate at $b = 2L$. Divide (3.14), minus one, by $N3^{-b}$. The positive leading ratio is $N^{2h-1} \rightarrow 0$, and the remainder ratio is $O(N^{1-2\log(5/3)}) \rightarrow 0$. Thus $A_N(2L) < 1$ eventually. The single crossing from Lemma 3.2 excludes all $b \leq 2L$. For $b \geq 2L$, the same enclosure gives

$$A_N(b) - 1 \leq 2^{-b} - (1 - \delta_N)N3^{-b}, \quad \delta_N = 2N(3/5)^{2L} \rightarrow 0. \quad (3.17)$$

The unrestricted maximum of its right side is $c\{(1 - \delta_N)N\}^{-p}$. Theorem 3.1 now proves (3.15).

This argument also localizes all near maximizers at the scale of the excess. Write $b = \hat{b}_N + d$. If $A_N(b) - 1 \geq (c - o(1))N^{-p}$, (3.17) first forces d into a compact interval: its normalized right side tends to zero as $d \rightarrow +\infty$, and is negative for sufficiently negative d . On compact d intervals,

$$N^p\{A_N(\hat{b}_N + d) - 1\} \rightarrow q^{-p}\{2^{-d} - q^{-1}3^{-d}\} \quad (3.18)$$

uniformly. The limiting function has its unique maximum at $d = 0$, so all global maximizers satisfy (3.16).

Convergence in (3.18) holds with two derivatives. Here is a direct justification for differentiating the remainder. The first derivative of the normalized gamma density inserts $\log t - \psi(b)$, and the second inserts $(\log t - \psi(b))^2 - \psi_1(b)$. Under a gamma law of rate m ,

$$\mathbb{E}(\log T - \psi(b))^2 = (\log m)^2 + \psi_1(b).$$

Therefore, by the positive bound for the Taylor remainder, its first two derivative errors are bounded by constants times $N^2(5^{-b} + 6^{-b})$, uniformly for large b . The explicit term $N4^{-b}$ has the same negligible derivative property. At $d = 0$, the limiting second derivative is

$$-Q, \quad Q = (\log 2)hq^{-p} > 0. \quad (3.19)$$

Every global maximizer eventually lies in a fixed neighborhood where the second derivative is strictly negative. Strict concavity there proves uniqueness. $\square$

3.3 A second asymptotic scale from the gamma threshold

The next Taylor moment in (3.14) is a useful bound, but it is not the next asymptotic term at the moving maximizing order. The correction comes from the layer $t \simeq (\log N)/2$. Define

$$\begin{aligned} \kappa &= \frac{1}{2} - \alpha + \alpha \log(2\alpha), & s &= \frac{1}{2} - \alpha, \\ D &= \sqrt{\frac{\alpha}{2\pi}} (2\alpha)^{-d_0} \Gamma(s). \end{aligned} \quad (3.20)$$

The constants satisfy $3/2 < \alpha < 5/2$, so $-2 < s < -1$ and $D > 0$. Elementary logarithm bounds give $p < \kappa < 2p - 1$. Their approximate values, used only for orientation, are

$$\begin{aligned} p &= 1.70951129135\dots, & \kappa &= 1.96959041447\dots, \\ c &= 0.16794779654\dots, & D &= 1.56777020071\dots \end{aligned}$$

Lemma 3.4 (Threshold-layer expansion). *Fix any $\alpha \in (3/2, 5/2)$, independently of its particular value in (3.13), and let $b = \alpha L + d$, $L = \log N$, with d in a compact interval. For the exact remainder in (3.14),*

$$\begin{aligned} \varepsilon_N(\alpha L + d) &= \sqrt{\frac{\alpha}{2\pi}} (2\alpha)^{-d} N^{-I(\alpha)} L^{-1/2} \\ &\times \left\{ \Gamma(s) + \frac{B_1(d)}{L} + O(L^{-2}) \right\}, \quad s = \frac{1}{2} - \alpha, \end{aligned} \quad (3.21)$$

where I is as in (3.12), and

$$B_1(d) = \left\{ \frac{d - d^2}{2\alpha} - \frac{1}{12\alpha} \right\} \Gamma(s) - (d - 1) \Gamma'(s) - \frac{\alpha}{2} \Gamma''(s). \quad (3.22)$$

The expansion is uniform with any fixed finite number of d derivatives. It extends to every finite order in L^{-1} .

Proof. Set

$$G_N(x) = \frac{(1-x)^{N-1}}{1+x}, \quad r_N(z) = G_N(z/N) - 1 + z \quad (0 \leq z \leq N).$$

Convexity and Lemma 3.2 imply the uniform bound

$$0 \leq r_N(z) \leq \min(z, z^2) \quad (N \geq 2). \quad (3.23)$$

Moreover $r_N(z) \rightarrow r(z) := e^{-z} - 1 + z$ on compact z intervals. The substitution $z = Ne^{-2t}$ in the exact gamma integral gives

$$\begin{aligned} \varepsilon_N(b) &= \frac{N^{-1/2}(L/2)^{b-1}}{2\Gamma(b)} \int_0^N z^{-1/2} \left(1 - \frac{\log z}{L} \right)^{b-1} \\ &\times \left(1 + \sqrt{\frac{z}{N}} \right) r_N(z) dz. \end{aligned} \quad (3.24)$$

The factor involving L tends to $z^{-\alpha}$. The limiting integral is

$$\int_0^\infty z^{-\alpha-1/2} (e^{-z} - 1 + z) dz = \Gamma(1/2 - \alpha). \quad (3.25)$$

Two integrations by parts prove this identity; both endpoint terms vanish because $3/2 < \alpha < 5/2$. In particular the individual divergent integrals of the three summands are never separated.

To justify convergence, choose $0 < \eta < \min(5/2 - \alpha, \alpha - 3/2)$. For large L , the factor in (3.24) is bounded by $z^{-\alpha-\eta}$ on $0 < z < 1$, and by $z^{-\alpha+\eta}$ on $1 < z < N$. Together with (3.23) and $1 + \sqrt{z/N} \leq 2$, these are integrable majorants at both endpoints. This proves the leading integral limit.

For the expansion put $w = \log z$ and split at $|w| \leq L^\theta$, where $\theta > 0$ is sufficiently small for the desired finite expansion order. The tails have integrable exponential envelopes in $|w|$, so their integrals are smaller than every inverse power of L . On the central interval,

$$(b-1)\log(1-w/L) + \alpha w = -\frac{(d-1)w + \alpha w^2/2}{L} + O\left(\frac{|w|^2 + |w|^3}{L^2}\right). \quad (3.26)$$

Removal of $\sqrt{z/N}$ and replacement of r_N by r contribute errors smaller than every inverse power of L . For the latter, the linear term cancels exactly; on the central interval expansion of $\log G_N(z/N)$ gives

$$|G_N(z/N) - e^{-z}| \leq CN^{-1}z^2e^{-z}$$

for large N . The logarithmic moments from (3.26) are the derivatives of the convergent Mellin integral (3.25). Taylor remainders are controlled by the same exponential envelopes with additional polynomial factors in $|w|$.

Stirling's expansion supplies

$$\begin{aligned} \frac{N^{-1/2}(L/2)^{b-1}}{2\Gamma(b)} &= \sqrt{\frac{\alpha}{2\pi}}(2\alpha)^{-d}N^{-I(\alpha)}L^{-1/2} \\ &\times \left[1 + \frac{1}{L} \left\{ \frac{d-d^2}{2\alpha} - \frac{1}{12\alpha} \right\} + O(L^{-2})\right]. \end{aligned} \quad (3.27)$$

Combining (3.26)–(3.27) proves (3.22). The argument works at every fixed finite order. Differentiation in d inserts powers of $\log(1-w/L)$ in the integral and logarithmic derivatives of the gamma prefactor. On $w < 0$, $\log(1+|w|/L) \leq |w|/L$. On $0 < w < L$, each fixed power of $|\log(1-w/L)|$ can be absorbed by decreasing the positive exponent $b-1$ by ηL : for $0 < u < 1$,

$$u^{\eta L} |\log u|^j \leq \left(\frac{j}{e\eta L}\right)^j \quad (j \geq 1).$$

Choosing the initial η with a further factor of two of room leaves the same integrable exponential tails. The gamma-prefactor derivatives remain bounded after normalization, by Stirling's differentiated expansion. These estimates apply in particular through two derivatives, and prove all the asserted uniform derivatives. $\square$

Theorem 3.5 (Second scale and displacement of the maximizing order). *With the constants in (3.13), (3.19), and (3.20),*

$$C_N = 1 + cN^{-p} + DN^{-\kappa}(\log N)^{-1/2}\{1 + O((\log N)^{-1})\}, \quad (3.28)$$

and the unique axis maximizing order satisfies

$$b_N = \hat{b}_N - \frac{D \log(2\alpha)}{Q} N^{-(\kappa-p)} (\log N)^{-1/2} \{1 + O((\log N)^{-1})\}. \quad (3.29)$$

More precisely, the braces in (3.28) begin $1 + B/\log N + O((\log N)^{-2})$, where

$$B = \frac{d_0 - d_0^2}{2\alpha} - \frac{1}{12\alpha} - (d_0 - 1)\psi(s) - \frac{\alpha}{2}\{\psi_1(s) + \psi(s)^2\}. \quad (3.30)$$

An expansion through every finite inverse power of $\log N$ at this second scale is obtained by the same Mellin-moment procedure.

Proof. At $b = \hat{b}_N$, Lemma 3.4 applies with $d = d_0$. The explicit term $N4^{-b}$ in (3.14) is $O(N^{-(2p-1)})$, smaller than $N^{-\kappa}(\log N)^{-m}$ for every fixed m . This proves the stated expansion first at $\hat{b}_N$.

For the optimizer, Theorem 3.3 already gives $b_N - \hat{b}_N = o(1)$ and uniform negative curvature of order N^{-p} . At $\hat{b}_N$ the derivative of the elementary leading pair vanishes. The derivative version of Lemma 3.4 shows that the remaining derivative is

$$-D \log(2\alpha) N^{-\kappa} L^{-1/2} \{1 + O(L^{-1})\}.$$

The leading second derivative is $-QN^{-p}$. The mean-value theorem first gives $b_N - \hat{b}_N = O(N^{-(\kappa-p)}L^{-1/2})$, and then substitution back into that derivative equation gives (3.29). The change in the maximum's value is $O(N^{-(2\kappa-p)}L^{-1})$, smaller than every $N^{-\kappa}L^{-m}$ at fixed m . The whole second-scale inverse-log expansion therefore transfers unchanged from $\hat{b}_N$ to the maximum. Division of (3.22) by $\Gamma(s)$ gives (3.30). $\square$

Remark 3.6 (A particularly late asymptotic regime). The inequalities $3/2 < \alpha < 5/2$ are essential. Here $5/2 - \alpha = 0.0336965\dots$ is small, and the lower endpoint of the Mellin integral decays slowly. The coefficient in (3.30) is correspondingly large in magnitude. The formulas are proved asymptotics as $N \rightarrow \infty$, without a claim that the second term is accurate at ordinary numerical truncation indices. In particular replacing it by the next Taylor moment $N^2 5^{-\hat{b}_N}/2$ would be incorrect: its exponent $\log 5/h - 2$ is strictly smaller than κ , and its ratio to the true second-scale correction grows without bound.

3.4 What sufficiently accurate near maximizers must do

Lemma 3.7 (A uniform small-outer-order tail). *Uniformly for $0 < a \leq 1$,*

$$J_N(a) = \frac{\sqrt{\pi}}{2^a \Gamma(a)} N^{-1/2} (\log N)^{a-1} \{1 + o(1)\}. \quad (3.31)$$

In particular, if $a \rightarrow 0$ and $a \log \log N \rightarrow 0$, then $J_N(a) \sim \sqrt{\pi} a N^{-1/2} / \log N$.

Proof. With $L = \log N$ and $z = Ne^{-2t}$,

$$J_N(a) = \frac{N^{-1/2}(L/2)^{a-1}}{2\Gamma(a)} \int_0^N z^{-1/2} \left(1 - \frac{\log z}{L}\right)^{a-1} f_N(z/N) dz.$$

On $0 < z \leq \sqrt{N}$, the factor raised to $a-1$ is bounded by one for $z < 1$ and by two for $z \geq 1$. Also $f_N(z/N) \leq e^{-z}$. The factor tends to one uniformly in $a \in (0, 1]$ at each fixed z , so dominated convergence gives $\Gamma(1/2)$ from this part, uniformly in a . The remaining part corresponds to $t < L/4$. For $0 < t < 1$, put $\rho = 1 - e^{-2t} < 1$ and use $(1 - e^{-2t})^N \leq 2t\rho^{N-1}$. For $1 < t < L/4$ use $\Psi_N(t) \leq e^{-\sqrt{N}}$. Since $1/\Gamma(a) = O(a)$ uniformly on $(0, 1]$, this part of $J_N(a)$ is $O(a\rho^{N-1} + ae^{-\sqrt{N}})$, negligible relative to the proposed main term uniformly in a . This proves (3.31). Finally $\Gamma(a)^{-1} \sim a$ and $(L/2)^a \rightarrow 1$ under the stated hypotheses. $\square$

Proposition 3.8 (Two-scale local profile in the order plane). *Suppose $a_N \geq 0$, $b_N > 0$, and*

$$\frac{a_N}{(\log N)N^{-(p-1/2)}} \rightarrow u \geq 0, \quad b_N - \hat{b}_N \rightarrow d \in \mathbb{R}.$$

Then

$$N^p \{R_N(a_N, b_N) - 1\} \rightarrow q^{-p} \{2^{-d} - q^{-1}3^{-d}\} - \sqrt{\pi}u. \quad (3.32)$$

Proof. The exact difference used in (3.7) has the form

$$R_N(a, b) - A_N(b) = -J_N(a)\zeta(b) + \Delta_N(a, b),$$

where

$$0 \leq \Delta_N(a, b) \leq (N+1)(1-3^{-a})\{\zeta(b) - 1 - 2^{-b}\}.$$

At $b = \hat{b}_N + O(1)$ the latter bound is $O(aN^{-p})$. Under the hypothesis $a_N \rightarrow 0$, so this is $o(N^{-p})$. Lemma 3.7 gives $N^p J_N(a_N) \rightarrow \sqrt{\pi} u$; the same conclusion is immediate at $a_N = 0$. Also $\zeta(b_N) \rightarrow 1$. Combine this with (3.18). $\square$

Corollary 3.9 (Quantitative approach to the maximizing boundary). *Suppose $a_N, b_N > 0$. The condition*

$$C_N - R_N(a_N, b_N) = o(N^{-p})$$

is equivalent to

$$a_N = o\left(N^{-(p-1/2)} \log N\right), \quad b_N - \hat{b}_N \rightarrow 0. \quad (3.33)$$

Proof. The hypothesis and Theorem 3.3 imply $R_N(a_N, b_N) > 1$ eventually. The exclusion argument in Theorem 3.1 forces $0 < a_N < 1$ and $b_N > 3 \log N/2$. Since $A_N(b_N) \leq C_N$, (3.9) gives the first conclusion. It also shows $A_N(b_N) - 1 \geq (c - o(1))N^{-p}$, so the localization proof in Theorem 3.3 gives the second. Conversely, (3.33) gives $u = d = 0$ in Proposition 3.8; its limit is c , which is also the limit of $N^p(C_N - 1)$. $\square$

The accuracy $o(N^{-p}) = o(C_N - 1)$ matters. An error merely $o(1)$ allows the broad moving-order phase region already present in the incoming report and need not force the outer order to zero. The results above prove eventual strict monotonicity through the exact axis reduction, while monotonicity for the entire sequence remains open. A theorem covering all finite truncation indices and an effective threshold for the eventual statements remain concrete further problems.

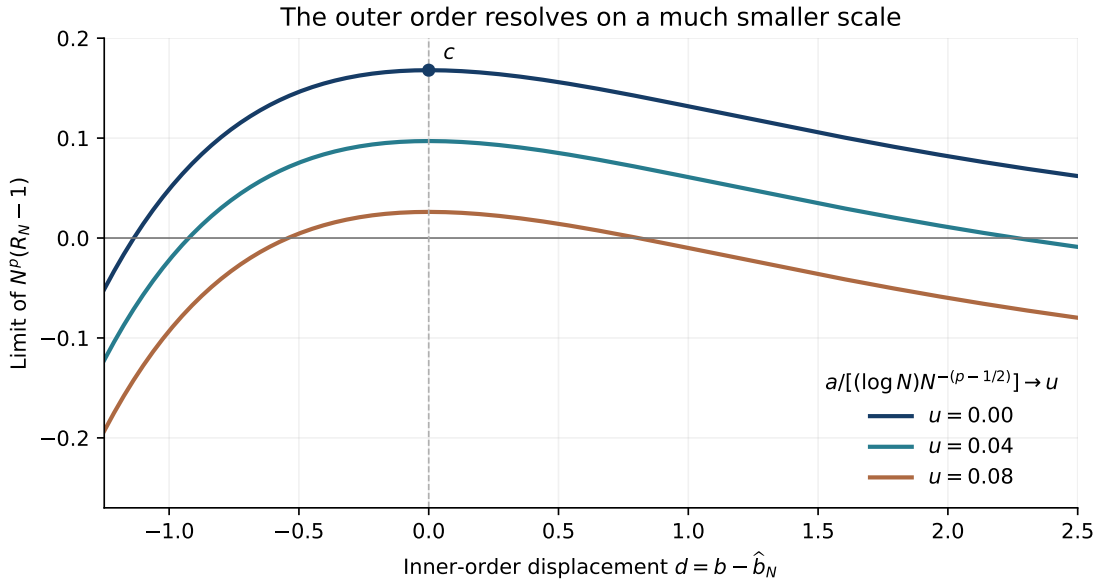


Figure 2: The exact limiting function in Proposition 3.8, at three values of the scaled outer order u . The maximum c occurs at $u = d = 0$. These curves depict the proved limit; they are not finite- N approximation error claims.

4 Polynomial Appell mesh and uniform Lerch zero thresholds

The incoming uniform-continuation article asks whether the roots of the reciprocal-gamma Appell polynomials admit a polynomial lower separation bound. Its estimate $(3n)^{-(n-2)}$ gives an effective zero theorem, but only a logarithmic growing-index range. We prove

$$\boxed{\text{mesh}(Q_n) \geq \frac{1}{10n^2} \quad (n \geq 2).} \quad (4.1)$$

A second improvement replaces a worst-distance estimate by a harmonic sum over the entire root configuration. Together these give a sufficient Lerch saturation threshold of order $n^4 \log^2 n$, and a relative location error independent of n . The constants below are explicit and deliberately conservative; no sharp mesh asymptotic is asserted.

4.1 Definitions and the strengthened zero theorem

Define the monic Appell polynomials by

$$\frac{e^{xt}}{\Gamma(1+t)} = \sum_{n \geq 0} Q_n(x) \frac{t^n}{n!}, \quad W_\rho(t) = \frac{1}{1 - \rho e^{-t}} \quad (0 \leq \rho \leq 1). \quad (4.2)$$

For $n \geq 0$, $k \geq 1$ and $a > 0$, put

$$F_{n,k}^\rho(a) = \int_0^\infty t^k e^{-at} W_\rho(t) Q_n(\log t) dt. \quad (4.3)$$

This is the manuscript's signed derivative family:

$$F_{n,k}^\rho = (-1)^{n+k} D_{n,k}^\rho, \quad D_{n,k}^\rho(a) = \begin{cases} \partial_a^k \sum_{m \geq 0} \frac{\rho^m \log^n(a+m)}{a+m}, & 0 \leq \rho < 1, \\ \gamma_n^{(k)}(a), & \rho = 1. \end{cases}$$

The endpoint $\rho = 1$ unifies the derivatives, not the generally divergent undifferentiated Lerch series. The Mellin derivation of (4.3) is already established in the manuscript. We recall below the positive-transform zero bound needed to make our localization complete.

Write the roots decreasingly as $x_{n,1} > \cdots > x_{n,n}$, set $c_{n,j} = e^{-x_{n,j}}$, and let $H_r = \sum_{h=1}^r h^{-1}$.

Theorem 4.1 (Polynomial threshold and simultaneous location). *For $n \geq 2$, define the explicit integer*

$$K_n = \left\lceil (16 + 240n^2 H_{n-1})^2 \right\rceil. \quad (4.4)$$

For every integer $k \geq K_n$ and every $\rho \in [0, 1]$, $F_{n,k}^\rho$ has exactly n positive zeros, all simple. Numbering them increasingly as $a_{n,j,k}^\rho$, one has

$$\boxed{\left| \log \frac{a_{n,j,k}^\rho}{k c_{n,j}} \right| < \frac{8}{\sqrt{k}}, \quad 1 \leq j \leq n.} \quad (4.5)$$

All assertions hold simultaneously over the entire deformation interval and over the smallest and largest zero; no compact restriction on the slopes $c_{n,j}$ is needed.

The proof is divided into an algebraic separation theorem and a scale-free analytic sign-transfer theorem.

4.2 Two distinct monotonicity statements for differential factors

For a real-rooted polynomial, its mesh is the minimum spacing of consecutive roots counted with multiplicity. Thus a multiple root gives mesh zero. In this subsection ∂ means differentiation in x .

Lemma 4.2 (Coordinatewise motion of ordered roots). *For a fixed real-rooted polynomial p , every ordered root of $p + cp'$, counted with multiplicity, is nonincreasing as $c > 0$ increases. Consequently each ordered root of*

$$P(x) = \prod_{j=1}^m (1 + c_j \partial) x^n$$

is nonincreasing in each positive parameter c_j separately.

Proof. Let the distinct roots of p be r_i , with multiplicities m_i . A root of multiplicity $m_i \geq 2$ is inherited with multiplicity $m_i - 1$. All other roots solve

$$S(x) := \sum_i \frac{m_i}{x - r_i} = -\frac{1}{c}.$$

There is one solution in each interval between consecutive distinct roots and one to the left of all roots. Since $S'(x) = -\sum_i m_i / (x - r_i)^2 < 0$, these solutions are simple, and

$$\frac{dx}{dc} = -\frac{1}{c^2 \sum_i m_i / (x - r_i)^2} < 0. \quad (4.6)$$

The inherited roots remain fixed. No moving root crosses an inherited one, because it remains in its designated complementary interval. This proves the first assertion and real-rootedness preservation. For the second assertion, hold all parameters except c_j fixed and apply the first to the real-rooted polynomial produced by the other factors. The factors commute. $\square$

Lemma 4.3 (Riesz mesh preservation). *For every real-rooted polynomial p and every real c ,*

$$\text{mesh}(p + cp') \geq \text{mesh}(p). \quad (4.7)$$

Proof. This classical theorem is recorded in [9, Theorem 1]. Here is the relevant argument. If $0 < h < \text{mesh}(p)$, then $p(x)$ and $p(x-h)$ have interlacing roots. The real-rooted-pencil criterion says that two real-rooted polynomials interlace if and only if every real linear combination is real-rooted or zero. To recall why, after common roots are removed, the partial fractions of their quotient have residues of one sign precisely in the interlacing case. Its imaginary part then has a fixed nonzero sign in the upper half-plane, so it takes no real value there. Conversely, if every real pencil member has real roots, the quotient takes no real value in that half-plane; continuity fixes the sign of its imaginary part, and approach to each pole forces the residues to have a common sign. This gives interlacing. Common roots can be restored, or treated by limits.

The operator $T = 1 + c\partial$ preserves real-rootedness and commutes with translation. Apply it to the interlacing pencil. Every real combination of $Tp(x)$ and $Tp(x-h)$ remains real-rooted or zero, so the shifted pair interlaces. For a polynomial and its translate by $h > 0$, this is equivalent to mesh at least h . Let h increase to $\text{mesh}(p)$. If the original mesh is zero, the assertion follows already from real-rootedness preservation. $\square$

Lemma 4.2 concerns the location of each root, whereas Lemma 4.3 compares a polynomial with its image under one operator. We do not claim that the mesh of $p + cp'$ is monotone as a function of c ; that stronger statement is false in general.

4.3 A Laguerre comparison with uniform spacing

The standard Laguerre coefficient formula, differential equation and three-term recurrence used here are collected in DLMF §§18.5, 18.8 and 18.9 [8]. We supply the bounds explicitly.

Lemma 4.4 (An equal-factor reference polynomial). *For every $n \geq 3$, the roots of*

$$(1 + \partial)^{3n} x^n = n! L_n^{(2n)}(-x) \quad (4.8)$$

are negative and simple. Every adjacent gap exceeds 3, and every root has absolute value less than $15n/2$.

Proof. More generally, for $m \geq n$, direct coefficient comparison gives

$$(1 + \partial)^m x^n = n! L_n^{(m-n)}(-x), \quad L_n^{(\alpha)}(z) = \sum_{r=0}^n (-1)^r \binom{n+\alpha}{n-r} \frac{z^r}{r!}.$$

Repeated real-root preservation removes all initial multiplicities after $n - 1$ nonzero factors. The resulting polynomial has strictly positive coefficients when $m \geq n$, so all its real roots are strictly negative. Equivalently the Laguerre roots are positive and simple.

Write $y = L_n^{(\alpha)}$. Its differential equation is

$$xy'' + (\alpha + 1 - x)y' + ny = 0.$$

The substitution $u = x^{(\alpha+1)/2} e^{-x/2} y$ gives

$$u'' + q(x)u = 0, \quad q(x) = \frac{2n + \alpha + 1}{2x} + \frac{1 - \alpha^2}{4x^2} - \frac{1}{4}. \quad (4.9)$$

For $\alpha = 2n$, maximizing the quadratic in $1/x$ yields

$$q(x) \leq \frac{12n^2 + 8n + 2}{16n^2 - 4} \leq 1 \quad (n \geq 3). \quad (4.10)$$

Indeed the difference of denominator and numerator is $4n^2 - 8n - 6 = 4(n-3)^2 + 16(n-3) + 6 > 0$. If $\lambda_i < \lambda_{i+1}$ are consecutive positive Laguerre roots, then u vanishes at both endpoints of this interval. Integration by parts and the Dirichlet Wirtinger inequality imply

$$\frac{\pi^2}{(\lambda_{i+1} - \lambda_i)^2} \int_{\lambda_i}^{\lambda_{i+1}} u^2 dx \leq \int_{\lambda_i}^{\lambda_{i+1}} (u')^2 dx = \int_{\lambda_i}^{\lambda_{i+1}} qu^2 dx \leq \int_{\lambda_i}^{\lambda_{i+1}} u^2 dx.$$

Thus every gap is at least $\pi > 3$.

For the largest root, the monic Laguerre recurrence identifies the zeros with the eigenvalues of the symmetric tridiagonal matrix having

$$J_{ii} = 2i - 1 + \alpha \quad (1 \leq i \leq n), \quad J_{i,i+1} = J_{i+1,i} = \sqrt{i(i+\alpha)}.$$

At $\alpha = 2n$, each diagonal is at most $4n - 1$, and each nonzero off-diagonal is less than $\sqrt{3}n$. The row-sum bound for eigenvalues therefore gives

$$\lambda_n \leq (4 + 2\sqrt{3})n - 1 < \frac{15n}{2},$$

where $\sqrt{3} < 7/4$. This proves both estimates. $\square$

4.4 Selecting a remote block from the gamma product

Theorem 4.5 (Polynomial reciprocal-gamma root separation). *For every integer $n \geq 2$, Q_n has n distinct real roots and satisfies (4.1).*

Proof. First take $n \geq 3$, and set $M = 15n^2$, $m = 3n$. Consider the finite block

$$B_n(x) = \prod_{j=M}^{M+m-1} (1 + \partial/j)x^n. \quad (4.11)$$

Let $r_1 < \dots < r_n < 0$ be the roots of the equal-factor polynomial in (4.8), and $R_1 < \dots < R_n$ those of B_n . Put $c_+ = 1/M$ and $c_- = 1/(M+m-1)$. Every coefficient of a differential factor in (4.11) lies between c_- and c_+ . Lemma 4.2, together with homogeneity of an equal-factor block, gives

$$c_+r_i \leq R_i \leq c_-r_i \quad (1 \leq i \leq n). \quad (4.12)$$

The inequalities have this orientation because all r_i are negative. Write $R_i = c_+r_i + \varepsilon_i$. By Lemma 4.4,

$$0 \leq \varepsilon_i < E, \quad E = \frac{45n^2}{2M^2},$$

since

$$(c_+ - c_-)\frac{15n}{2} < \frac{m}{M^2} \frac{15n}{2} = E.$$

The errors lie on the same side of the reference roots, so each gap loses at most E , not $2E$. Consequently

$$R_{i+1} - R_i > \frac{3}{M} - E = \frac{3}{15n^2} - \frac{45n^2}{2(15n^2)^2} = \frac{1}{10n^2}. \quad (4.13)$$

The Weierstrass product for the reciprocal gamma function yields the finite-product Appell polynomials

$$Q_{n,N}(x) = \prod_{j=1}^N (1 + \partial/j)(x + \gamma - H_N)^n.$$

For every $N \geq M+m-1$, start with the block B_n , restore the remaining factors, and translate by $\gamma - H_N$. All factors and translations commute. Lemma 4.3 shows that the restored factors cannot decrease mesh; translations preserve it. Thus $\text{mesh}(Q_{n,N}) > 1/(10n^2)$. The locally uniform convergence of the Weierstrass product implies coefficientwise convergence $Q_{n,N} \rightarrow Q_n$. Ordered real roots of monic fixed-degree polynomials vary continuously with the coefficients, so the non-strict gap bound passes to the limit. It proves simplicity as well as real-rootedness of Q_n .

Finally $Q_2(x) = (x + \gamma)^2 - \zeta(2)$ has $\text{mesh } 2\sqrt{\zeta(2)} > 2 > 1/40$, which handles the omitted index. $\square$

The choice of a distant block is useful because its coefficients $1/j$ vary little across $3n$ successive factors. It supplies separated roots before any potentially troublesome infinite-product limit is taken. The remaining gamma factors then preserve this separation.

4.5 A scale-free sign-transfer estimate

Let Y_k have the gamma density

$$\frac{k^{k+1}}{k!} y^k e^{-ky}, \quad y > 0. \quad (4.14)$$

Its shape is $k+1$ and rate is k .

Lemma 4.6 (An exponential log-moment bound). *For positive integers m, k with $k \geq 16m^2$,*

$$\mathbb{E}(e^{m|\log Y_k|} - 1) \leq \sqrt{2(e^{4m^2/k} - 1)} \leq \sqrt{2/3} < 1. \quad (4.15)$$

Proof. The square of $e^{m|\log y|} - 1$ is either $(y^m - 1)^2$ or $(y^{-m} - 1)^2$, hence is at most their sum. Cauchy–Schwarz bounds the square of the left side of (4.15) by

$$\mathbb{E}Y_k^{2m} + \mathbb{E}Y_k^{-2m} - 2\mathbb{E}Y_k^m - 2\mathbb{E}Y_k^{-m} + 2.$$

The integer moments are

$$\mathbb{E}Y_k^r = \prod_{j=1}^r (1 + j/k), \quad \mathbb{E}Y_k^{-r} = \prod_{j=0}^{r-1} (1 - j/k)^{-1} \quad (r \leq k).$$

Both products are at least one. Since $k \geq 4m$, logarithmic bounds give

$$\log \mathbb{E}Y_k^{2m} \leq \frac{m(2m+1)}{k} \leq \frac{3m^2}{k}, \quad \log \mathbb{E}Y_k^{-2m} \leq \frac{2}{k} \sum_{j=0}^{2m-1} j \leq \frac{4m^2}{k}.$$

The second estimate uses $-\log(1-u) \leq 2u$ for $0 \leq u \leq 1/2$. This proves the first claimed inequality. Finally $4m^2/k \leq 1/4$ and $e^{1/4} \leq (1 - 1/4)^{-1} = 4/3$ prove the second. $\square$

Proposition 4.7 (Root-distance sign transfer). *For real s not a root of Q_n , set*

$$S_n(s) = \sum_{\ell=1}^n \frac{1}{|s - x_{n,\ell}|}, \quad m = \lceil 1 + S_n(s) \rceil. \quad (4.16)$$

If $k \geq 16m^2$, then simultaneously for $\rho \in [0, 1]$,

$$\left| \frac{(ke^{-s})^{k+1} F_{n,k}^\rho(ke^{-s})}{k! W_\rho(e^s) Q_n(s)} - 1 \right| \leq \sqrt{2/3} < 1. \quad (4.17)$$

In particular $F_{n,k}^\rho(ke^{-s})$ and $Q_n(s)$ have the same sign.

Proof. A direct substitution in (4.3) gives the exact identity

$$\frac{(ke^{-s})^{k+1} F_{n,k}^\rho(ke^{-s})}{k! W_\rho(e^s) Q_n(s)} = \mathbb{E} \left[\frac{W_\rho(e^s Y_k)}{W_\rho(e^s)} \frac{Q_n(s + \log Y_k)}{Q_n(s)} \right]. \quad (4.18)$$

The logarithmic derivative of the weight satisfies

$$\left| \frac{d \log W_\rho(x)}{d \log x} \right| = \frac{x\rho}{e^x - \rho} \leq 1.$$

Indeed $e^x - \rho \geq 1 + x - \rho \geq \rho x$. This includes both endpoints of the deformation interval. For $R = W_\rho(e^{s+u})/W_\rho(e^s)$ one obtains $R \leq e^{|u|}$ and $|R - 1| \leq e^{|u|} - 1$. The product over polynomial roots gives

$$\left| \frac{Q_n(s+u)}{Q_n(s)} - 1 \right| \leq \prod_{\ell=1}^n \left(1 + \frac{|u|}{|s - x_{n,\ell}|} \right) - 1 \leq e^{S_n(s)|u|} - 1.$$

Combining the two estimates yields

$$\left| R \frac{Q_n(s+u)}{Q_n(s)} - 1 \right| \leq e^{(1+S_n(s))|u|} - 1 \leq e^{m|u|} - 1.$$

Take expectations in (4.18) and apply Lemma 4.6. These estimates also give the needed absolute integrability. Since the normalized quantity lies within distance less than one of 1, it is positive. $\square$

4.6 Harmonic root distances and completeness of localization

Lemma 4.8 (Positive-transform zero bound). *For $n \geq 0$, $k \geq 1$ and $0 \leq \rho \leq 1$, the function $F_{n,k}^\rho$ has at most n positive zeros, counted with multiplicity.*

Proof. A real kernel with N sign changes has a Laplace transform with at most N positive zeros, counted with multiplicity, when its exponentially weighted polynomial moments converge. One direct proof is induction. At a sign-change point t_0 , multiplication of the kernel by $t_0 - t$ removes exactly that sign change. Its transform is

$$(\partial_a + t_0)F(a) = e^{-t_0 a} \frac{d}{da} (e^{t_0 a} F(a)).$$

Any finite collection of zeros of F with total multiplicity M forces at least $M - 1$ zeros of the transformed function by Rolle's theorem with multiplicities. The induction hypothesis gives $M \leq N$. The fixed-sign case begins the induction, and the same induction precludes an identically zero transform.

Here $Q_n(\log t)$ has n simple sign changes, whereas $t^k W_\rho(t) > 0$. The bound $W_\rho(t) \leq W_1(t) \leq 1 + t^{-1}$ ensures integrability at zero, and e^{-at} supplies it at infinity, also after any fixed number of a -derivatives. The variation argument therefore applies. $\square$

Theorem 4.9 (Saturation from any mesh bound). *Suppose $n \geq 2$ and $\delta > 0$ is a proved lower bound for $\text{mesh}(Q_n)$. If*

$$\sqrt{k} \geq 16 + \frac{24H_{n-1}}{\delta}, \quad (4.19)$$

then the full zero-count and location conclusions of Theorem 4.1 hold.

Proof. Set $\eta = 8/\sqrt{k}$. Since $H_{n-1} \geq 1$, (4.19) implies $\eta \leq \delta/3$. At either endpoint $s = x_{n,j} \pm \eta$, the distinguished root is at distance η , while every other root obeys

$$|s - x_{n,\ell}| \geq |\ell - j|\delta - \eta \geq \frac{2}{3}|\ell - j|\delta.$$

Consequently

$$\begin{aligned} S_n(s) &\leq \frac{1}{\eta} + \frac{3}{2\delta}(H_{j-1} + H_{n-j}) \\ &\leq \frac{\sqrt{k}}{8} + \frac{3H_{n-1}}{\delta}. \end{aligned} \quad (4.20)$$

Thus

$$m = \lceil 1 + S_n(s) \rceil \leq 2 + \frac{\sqrt{k}}{8} + \frac{3H_{n-1}}{\delta} \leq \frac{\sqrt{k}}{4},$$

where the last inequality is exactly (4.19). Proposition 4.7 transfers the opposite polynomial signs at both endpoints to the corresponding values of $F_{n,k}^\rho$, uniformly in ρ .

The n intervals $[x_{n,j} - \eta, x_{n,j} + \eta]$ are disjoint, because $2\eta \leq 2\delta/3 < \delta$. The decreasing coordinate change $a = ke^{-s}$ gives n disjoint positive intervals, each with opposite endpoint signs, hence each containing a zero. By Lemma 4.8 these are all the positive zeros; each interval contains exactly one, and every zero is simple. The interval endpoints give (4.5), with a strict inequality because the transferred endpoint signs are nonzero. $\square$

Proof of Theorem 4.1. Insert the proved value $\delta = 1/(10n^2)$ from Theorem 4.5 into Theorem 4.9. $\square$

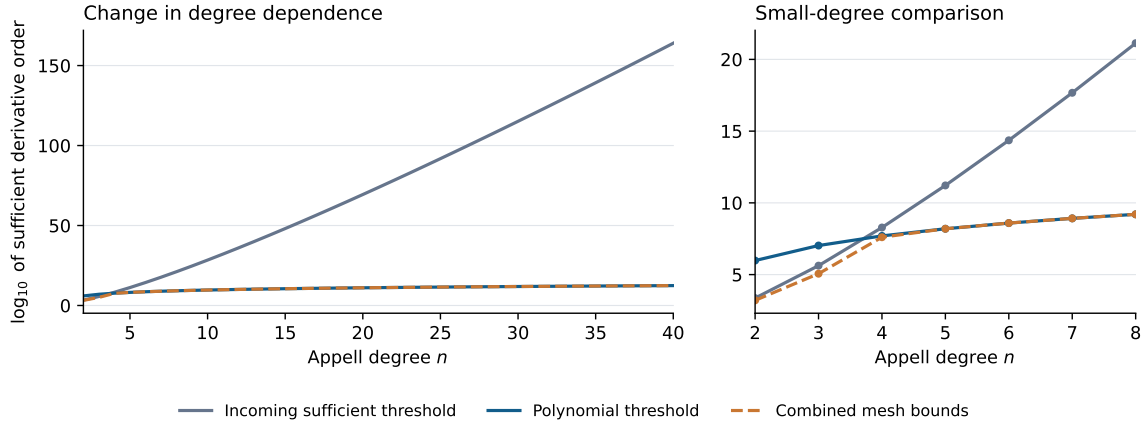


Figure 3: Comparison of proved sufficient derivative-order thresholds. The incoming threshold is $576n^2(3n)^{2n-4}$. The polynomial threshold is (4.4), and the combined bound uses δ_n^{comb} in the improved transfer theorem. The ordinate is the base-ten logarithm of each integer threshold; no numerical zero search is involved.

Combining proved mesh bounds at small degrees. The incoming bound $\text{mesh}(Q_n) \geq (3n)^{-(n-2)}$ [4, Section 4] and Theorem 4.5 may both be retained. Set

$$\delta_n^{\text{comb}} = \max\{(3n)^{-(n-2)}, 1/(10n^2)\}, \quad K_n^{\text{comb}} = \left\lceil \left(16 + \frac{24H_{n-1}}{\delta_n^{\text{comb}}}\right)^2 \right\rceil.$$

Theorem 4.9 applies with this bound too. It uses the better old separation estimate at $n = 2, 3, 4$ and the polynomial estimate thereafter, while keeping the improved $8/\sqrt{k}$ location error throughout. These remain sufficient thresholds, not assertions about the first index at which the zero count is saturated.

4.7 A polynomial growing-index regime

Corollary 4.10 (Joint growth of index and derivative order). *Fix $0 < \alpha < 1/\sqrt{60}$. For all sufficiently large integers k , simultaneously for*

$$2 \leq n \leq \alpha \frac{k^{1/4}}{\sqrt{\log k}}, \quad 1 \leq j \leq n, \quad 0 \leq \rho \leq 1, \quad (4.21)$$

the family has exactly n simple positive zeros and satisfies (4.5). In particular

$$\frac{a_{n,j,k}^\rho}{ke^{-x_{n,j}}} = 1 + O(k^{-1/2})$$

uniformly throughout this joint range. The same conclusion holds for any range $n = o(k^{1/4}/\sqrt{\log k})$.

Proof. The required square root of the threshold is $16 + 240n^2H_{n-1}$, which increases with n . At the upper endpoint of (4.21),

$$H_{n-1} = \left(\frac{1}{4} + o(1)\right) \log k, \quad \frac{16 + 240n^2H_{n-1}}{\sqrt{k}} \leq 60\alpha^2 + o(1) < 1.$$

The threshold condition therefore holds for all smaller n too. Theorem 4.1 applies, and exponentiating its uniform logarithmic error gives the stated relative error. $\square$

Remark 4.11 (Scope of the improvement). Theorem 4.5 settles the incoming question asking for any polynomial Appell mesh bound, with exponent 2 and an explicit constant. The harmonic-distance estimate also improves the transfer argument itself. These results do not identify the actual smallest-gap asymptotic, the sharp saturation threshold at a fixed index, or uniform constant-offset and all-orders zero expansions when n grows. The existing fixed-index expansion remains compatible with the stronger leading-order joint theorem proved here.

5 Two exact Gaussian sign transitions below outer order minus one

The incoming Euler moment identities prove $g_{s,b} < 0$ for every $s \geq -1$ and $b > 0$ [2, Section 6]. They also exhibit opposite signs farther down the outer-order ladder and ask for the resulting sign changes. We resolve the first two integer cases completely in the inner order b .

5.1 A finite identity at every negative integer outer order

At $a = 0$, (2.1) simplifies to

$$F_{0,b}(z) = \frac{z \operatorname{Li}_b(z)}{1-z} = \operatorname{Li}_0(z) \operatorname{Li}_b(z).$$

The product rule and $\mathcal{D} \operatorname{Li}_s(z) = \operatorname{Li}_{s-1}(z)$ give the following exact identity, with no relation search.

Proposition 5.1 (Negative-order product formula). *For every integer $m \geq 0$ and every $b > 0$,*

$$F_{-m,b}(z) = \sum_{j=0}^m \binom{m}{j} \operatorname{Li}_{-j}(z) \operatorname{Li}_{b-m+j}(z) \quad (5.1)$$

on the principal slit plane. In particular the Gaussian value is a finite combination of shifted Dirichlet beta and eta values.

Proof. In the disk $F_{-m,b} = \mathcal{D}^m F_{0,b}$, by termwise differentiation. Apply the finite Leibniz rule to the displayed product, then continue analytically. No convergence of the original series at $z = i$ is asserted or needed. $\square$

For all complex s , analytic continuation of the residue-class identity gives

$$\operatorname{Li}_s(i) = -2^{-s} \eta(s) + i\beta(s). \quad (5.2)$$

Using $\operatorname{Li}_0(i) = (-1 + i)/2$, $\operatorname{Li}_{-1}(i) = -1/2$, $\operatorname{Li}_{-2}(i) = -i/2$, and $\operatorname{Li}_{-3}(i) = 1$ in (5.1) gives

$$2g_{-2,b} = -\beta(b-2) - 2\beta(b-1) - 2^{2-b} \eta(b-2) + 2^{-b} \eta(b), \quad (5.3)$$

$$2g_{-3,b} = 2\beta(b) - \beta(b-3) - 3\beta(b-2) - 2^{3-b} \eta(b-3) + 3 \cdot 2^{1-b} \eta(b-1). \quad (5.4)$$

The shifted arguments in these formulas can be nonpositive; they have their continued values. The next theorem is a global sign assertion about these finite combinations.

5.2 Rational kernels and one-crossing Mellin integrals

Lemma 5.2 (A Mellin sign-change principle). *Suppose a real function w is positive on $(0, \tau)$ and negative on (τ, ∞) , up to null sets, and*

$$J(b) = \int_0^\infty t^{b-1} w(t) dt$$

and its first logarithmic derivative integral converge locally uniformly for $b > 0$. Then $\tau^{-b}J(b)$ is strictly decreasing on $(0, \infty)$. Consequently J has at most one positive zero. If it has one, that zero is simple and J' is negative there.

Proof. Differentiation gives

$$\frac{d}{db}\{\tau^{-b}J(b)\} = \tau^{-b} \int_0^\infty t^{b-1} \log(t/\tau) w(t) dt < 0.$$

The integrand is negative on both sides of τ . At a zero of J , the same inequality gives $J' < 0$. $\square$

Theorem 5.3 (Complete first negative-order sign transitions). *For each $m \in \{2, 3\}$ there is exactly one $b_m > 0$ such that $g_{-m, b_m} = 0$. This zero is simple, and*

$$g_{-m, b} > 0 \quad (0 < b < b_m), \quad g_{-m, b} < 0 \quad (b > b_m). \quad (5.5)$$

Moreover $0 < b_2 < 1 < b_3$. The two Dirichlet combinations (5.3) and (5.4) have exactly the same respective zero and sign descriptions.

Proof. Differentiate the axis case of (2.1) under its bounded rational integrand. If $Y_b = e^{-T_b}$ and $T_b \sim \text{Gamma}(b, 1)$, then

$$g_{-m, b} = \mathbb{E}q_m(Y_b), \quad q_m(y) = \text{Im} \left[\mathcal{D}^m \frac{z^2}{(1-z)(1-zy)} \right]_{z=i}.$$

For each fixed m , this differentiation is justified uniformly on a complex neighborhood of i , because $0 \leq y \leq 1$ and the denominators stay away from zero. Direct differentiation yields

$$q_2(y) = \frac{(1+y)(y^4 + 6y^2 - 3)}{2(1+y^2)^3}, \quad (5.6)$$

$$q_3(y) = -\frac{(1+y)(y^4 - 22y^2 + 1)}{(1+y^2)^4}. \quad (5.7)$$

Each rational function has exactly one zero in $(0, 1)$, namely

$$r_2 = \sqrt{-3 + 2\sqrt{3}}, \quad r_3 = \sqrt{11 - 2\sqrt{30}}.$$

It is negative below that zero and positive above it. After the change $y = e^{-t}$, write

$$\Gamma(b)g_{-m, b} = \int_0^\infty t^{b-1} e^{-t} q_m(e^{-t}) dt. \quad (5.8)$$

The kernel $w_m(t) = e^{-t} q_m(e^{-t})$ is positive before $\tau_m = -\log r_m$ and negative after it. It is bounded at zero and decays exponentially at infinity, so all local parameter differentiations are legitimate. Lemma 5.2 proves at most one simple crossing, from positive to negative.

As $b \downarrow 0$, $T_b \rightarrow 0$ in probability because $\mathbb{E}T_b = b$. As $b \rightarrow \infty$, $T_b \rightarrow \infty$ in probability, for example by Chebyshev's inequality using its mean b and variance b . Since q_m is bounded and continuous on $[0, 1]$, these limits give

$$\begin{aligned} g_{-2, 0+} &= q_2(1) = \frac{1}{2}, & g_{-2, \infty} &= q_2(0) = -\frac{3}{2}, \\ g_{-3, 0+} &= q_3(1) = \frac{5}{2}, & g_{-3, \infty} &= q_3(0) = -1. \end{aligned}$$

Continuity therefore proves existence. Finally the exact values

$$g_{-2, 1} = -\frac{3}{4} + \frac{\log 2}{4} < 0, \quad g_{-3, 1} = 1 + \frac{\pi}{4} > 0$$

locate the two roots relative to one. They follow either from (5.3)–(5.4) or from differentiating $F_{1,1}(z) = \frac{1}{2} \log^2(1-z)$. These two isolated evaluations were already displayed in [2, Section 6]; the new conclusion is the complete one-crossing classification for all $b > 0$. $\square$

Independent numerical evaluation of the Dirichlet formulas and the Mellin integrals gives

$$b_2 = 0.4238611967707581269539978159\dots, \quad b_3 = 2.4369081043976327447535806729\dots$$

These decimals are diagnostics; uniqueness, simplicity, and the sign classification are analytic results. No small-residual test is used to decide whether a zero exists.

Why the next integer orders need new analysis. The rational kernel construction and finite identity (5.1) work for every m . A one-crossing argument does not automatically persist. For example,

$$q_5(y) = -\frac{(1+y)(y^8 - 236y^6 + 1446y^4 - 236y^2 + 1)}{(1+y^2)^6}$$

has two zeros in $(0, 1)$. To see this exactly, divide its degree-eight numerator by y^4 and set $T = y^2 + y^{-2} > 2$. The resulting quadratic is $T^2 - 236T + 1444$, whose distinct roots $118 \pm 8\sqrt{195}$ both exceed 2. The map $y \mapsto T$ decreases bijectively from $(0, 1)$ onto $(2, \infty)$, giving exactly two simple zeros. More general Mellin variation bounds give an upper bound by the number of kernel sign changes, but existence of all the permitted crossings is a separate problem. This is a concrete continuation of the negative-order question, not a proposed blanket extension of Gaussian negativity.

6 Exact sampling of the unresolved mixed Gaussian generator

The mixed Gaussian sums

$$S_p = \sum_{n=0}^{\infty} \frac{(-1)^n H_n}{(2n+1)^p}$$

have a proved reduction at $p = 2, 4$, while the fixed-weight reductions at $p = 6, 8$ remain conjectural in the pinned manuscript. The incoming uniform-continuation report additionally supplies conjectures at $p = 10, 12$. The following results do not settle these fixed-weight questions. They give an exact arithmetic sampling theorem for the whole generator, explicit finite parts at its poles, and convergent identities involving the even indices. A new, separately validated candidate at $p = 14$ follows below.

Write $L = \log 2$, $G = \beta(2)$ and

$$\mathcal{S}(t) = \sum_{p \geq 1} S_p t^{p-1}, \quad \mathcal{O}(t) = \frac{\mathcal{S}(t) - \mathcal{S}(-t)}{2} = \sum_{m \geq 1} S_{2m} t^{2m-1}.$$

The canonical manuscript already proves the meromorphic continuation of $\mathcal{S}$, its Mellin integral and its simple poles. We retain these facts as prerequisites, and include the short derivations needed for the new formulas. In particular, no new assertion of period independence or of minimal depth is made.

6.1 A shift equation and rational sampling with all parameter jets

Put

$$B(t) = \int_0^1 \frac{x^{-t}}{1+x^2} dx = \frac{1}{4} \left\{ \psi\left(\frac{3-t}{4}\right) - \psi\left(\frac{1-t}{4}\right) \right\}, \quad \Re t < 1,$$

and continue B meromorphically. Its elementary recurrence and reflection formula are

$$B(t) + B(t-2) = \frac{1}{1-t}, \quad B(t) + B(-t) = \frac{\pi}{2 \cos(\pi t/2)}. \quad (6.1)$$

The first follows by adding the two integrands; the second follows by splitting $\int_0^\infty x^{-t}/(1+x^2) dx$ at one.

Theorem 6.1 (Meromorphic shift equation). *On the complex plane, as an identity of meromorphic functions,*

$$\mathcal{S}(t) + \mathcal{S}(t-2) = \frac{2B(t-2) - L}{1-t}. \quad (6.2)$$

Removable values and finite parts are taken after combining the terms.

Proof. Let $q(x) = \log(1+x^2)/(1+x^2)$. The defining generating integral is $\mathcal{S}(t) = -\int_0^1 x^{-t} q(x) dx$ for $\Re t < 3$. Adding its shift by two gives $-\int_0^1 x^{-t} \log(1+x^2) dx$. For $t \neq 1$, integration by parts gives precisely (6.2); the lower boundary term vanishes because $x^{1-t} \log(1+x^2) = O(x^{3-\Re t})$. The identity first holds in an open half-plane and extends by the meromorphic identity theorem. The point $t = 1$ is removable on both sides. $\square$

We use the series convention

$$\mathrm{Li}_{a,b}(x, y) = \sum_{n>m \geq 1} \frac{x^n y^m}{n^a m^b}.$$

For $d \geq 1$ let $\mathcal{R}_d = \{\xi : \xi^d = -1\}$. No member of $\mathcal{R}_d$ equals one. At unit-modulus arguments, we use the radial boundary values of the multiple polylogarithms; all values below also have ordinarily convergent defining series. Indeed, for $r \geq 1$ the double series is absolutely convergent. For $r = 0$, write its inner argument as $y = \xi/\eta$ and outer argument as $x = \xi^{-1} \neq 1$. If $y \neq 1$, the inner sum is $\mathrm{Li}_1(y) + O(n^{-1})$, so Dirichlet's test treats the main outer series and the remainder converges absolutely. If $y = 1$, H_{n-1}/n decreases eventually to zero, and the same test applies.

Theorem 6.2 (Cyclotomic rational sampling and parameter jets). *Let $1 \leq A \leq d$ be integers, put $\alpha = 1 - 2A/d$, and let $r \geq 0$ be an integer. Then*

$$\frac{\mathcal{S}^{(r)}(\alpha)}{r!} = \frac{1}{2} \left(\frac{d}{2}\right)^r \sum_{\xi, \eta \in \mathcal{R}_d} \xi^A \mathrm{Li}_{r+1,1} \left(\xi^{-1}, \frac{\xi}{\eta} \right). \quad (6.3)$$

$$\frac{\mathcal{S}^{(r)}(\alpha)}{r!} = \sum_{k \geq 0} \binom{k+r}{r} S_{k+r+1} \alpha^k. \quad (6.4)$$

Thus every rational sample in $[-1, 1)$ and every parameter jet is an explicit $\mathbb{Q}(\zeta_{2d})$ -linear combination of cyclotomic doubles of the fixed weight $r+2$. All sums on the right of (6.3) are finite; complex-conjugate terms make the result real. Choosing $A/d = (1-\alpha)/2$ in lowest terms gives level $2d$ as an upper bound, not a claim of minimal conductor. The shift equation extends these evaluations to every rational sample at which $\mathcal{S}$ is regular, and gives Laurent coefficients at its remaining rational points.

Proof. Substitute $u = x^2$ in the Mellin integral and then $u = v^d$. This gives

$$\mathcal{S}(\alpha) = -\frac{d}{2} \int_0^1 \frac{v^{A-1} \log(1+v^d)}{1+v^d} dv.$$

The exact partial fraction identity is

$$\frac{d v^{A-1}}{1+v^d} = - \sum_{\xi \in \mathcal{R}_d} \frac{\xi^A}{v - \xi}. \quad (6.5)$$

Both sides are proper rational functions with the same simple poles and residues: the residue on the left at ξ is $\xi^{A-d} = -\xi^A$. Their difference therefore vanishes. Along $0 \leq v \leq 1$,

$$\log(1+v^d) = \sum_{\eta \in \mathcal{R}_d} \log(1-v/\eta).$$

The logarithms are their principal values, continued from $v = 0$; equality follows by differentiating and matching the value at zero. None of the factors meets its branch cut on this path.

Differentiating r times with respect to t inserts $(-\log x)^r = (d/2)^r (-\log v)^r$. Finally, the standard convergent iterated-integral formula, with the displayed series convention, is

$$\text{Li}_{r+1,1}(\xi^{-1}, \xi/\eta) = \frac{1}{r!} \int_0^1 \frac{(-\log v)^r \log(1 - v/\eta)}{v - \xi} dv.$$

Substitution proves (6.3). All integrals converge absolutely; at zero the logarithm is $O(v)$, and the root denominators do not vanish on $[0, 1]$. Parameter differentiation is justified on compact subsets of $\Re t < 3$ by integrable powers of $|\log x|$. Taylor differentiation inside $|t| < 3$ proves (6.4). Repeated application and differentiation of (6.2) proves the last assertion. $\square$

For example, at $\alpha = 1/3$ one may take $(A, d) = (1, 3)$, so the whole infinite sum $\sum_{p \geq 1} S_p 3^{1-p}$ is a finite combination of weight-two values at sixth roots of unity. At $\alpha = 1/2$, one takes $(A, d) = (1, 4)$ and obtains level eight. This is a fixed-weight evaluation of an infinite generating sum; it is not a termwise reduction of the individual periods S_p .

6.2 Elementary values and finite parts of the odd generator

Define the rational finite sums

$$a_N = \sum_{j=1}^N \frac{(-1)^{j-1}}{2j-1}, \quad b_N = \sum_{j=1}^N \frac{a_{j-1}}{2j-1}, \quad c_N = \sum_{j=1}^N \frac{(-1)^{j-1}}{(2j-1)^2},$$

with empty sums zero. Also put

$$\bar{H}_N^{(s)} = \sum_{j=1}^N \frac{(-1)^{j-1}}{j^s}, \quad e_N = \sum_{j=1}^N \frac{\bar{H}_{j-1}^{(1)}}{j}, \quad P = \frac{L^2}{4} - \frac{\pi^2}{48}.$$

The notation $\text{FP}_{t=t_0}$ denotes the constant term in the Laurent expansion in the local coordinate $t - t_0$.

Theorem 6.3 (All integer samples of the unresolved parity component). *For every integer $N \geq 0$,*

$$\boxed{\mathcal{O}(2N) = (-1)^N \{c_N + 2b_N - a_N L\}.} \quad (6.6)$$

At $t = 2N + 1$, its residue is $(-1)^{N+1} H_N/2$, and its finite part is

$$\begin{aligned} \text{FP}_{t=2N+1} \mathcal{O}(t) = (-1)^N \left\{ P - \frac{L}{2} (H_N + \bar{H}_N^{(1)}) \right. \\ \left. + \frac{H_N^{(2)} + \bar{H}_N^{(2)}}{4} + \frac{e_N}{2} \right\}. \end{aligned} \quad (6.7)$$

For $N = 0$ the pole is absent, and (6.7) is the ordinary value $\mathcal{O}(1) = P$.

Proof. First, the elementary beta/reflection evaluation gives $\mathcal{S}(0) = G - \pi L/2$. Direct integration after $u = x^2$ gives

$$\mathcal{S}(-1) = -\frac{L^2}{4}, \quad \mathcal{S}(1) = \frac{L^2}{4} - \frac{\pi^2}{24}.$$

The recurrence (6.1), starting with $B(0) = \pi/4$, yields

$$B(2j) = (-1)^j (\pi/4 + a_j), \quad B(-2j) = (-1)^j (\pi/4 - a_j).$$

Let $h_N^o = \sum_{j=1}^N (2j-1)^{-1}$. Iterating (6.2) at positive and negative even integers gives

$$\begin{aligned}\mathcal{S}(2N) &= (-1)^N \{\mathcal{S}(0) - La_N + (\pi/2)h_N^o + 2b_N\}, \\ \mathcal{S}(-2N) &= (-1)^N \{\mathcal{S}(0) + La_N + (\pi/2)h_N^o - 2b_N - 2c_N\}.\end{aligned}$$

Subtracting proves (6.6).

For the odd finite parts, the normally convergent pole expansion of B or its digamma formula gives, for $j \geq 0$,

$$\text{Res}_{t=2j+1} B(t) = (-1)^{j+1}, \quad \text{FP}_{t=2j+1} B(t) = \frac{(-1)^j}{2} (\bar{H}_j^{(1)} - L).$$

Write $\mathcal{C}_N^{\text{fp}} = \text{FP}_{t=2N+1} \mathcal{S}(t)$. Extracting the constant term of (6.2) gives, for $N \geq 1$,

$$\mathcal{C}_N^{\text{fp}} + \mathcal{C}_{N-1}^{\text{fp}} = \frac{(-1)^N}{2N^2} + \frac{(-1)^N \bar{H}_{N-1}^{(1)}}{2N} + \frac{1 - (-1)^N}{2N} L.$$

The first term is essential: it comes from the pole of $B(t-2)$ times the linear term of $(1-t)^{-1}$. Starting with $\mathcal{C}_0^{\text{fp}} = L^2/4 - \pi^2/24$, the recurrence gives

$$\mathcal{C}_N^{\text{fp}} = (-1)^N \left\{ \mathcal{C}_0^{\text{fp}} + \frac{H_N^{(2)}}{2} + \frac{e_N}{2} - \frac{L}{2} (H_N + \bar{H}_N^{(1)}) \right\}.$$

On the negative odd axis, polynomial division of $u^N/(1+u)$ gives

$$\mathcal{S}(-(2N+1)) = (-1)^N \left\{ -\frac{L^2}{4} + \frac{L}{2} (H_N + \bar{H}_N^{(1)}) - \frac{e_N + \bar{H}_N^{(2)}}{2} \right\}.$$

Taking half the difference proves (6.7). The residue is half the already known residue of $\mathcal{S}$ at the positive pole; $\mathcal{S}(-t)$ is regular there. $\square$

6.3 Convergent identities after subtracting known poles

Corollary 6.4 (Three explicit identities across all even indices). *The following series converge absolutely and satisfy*

$$\sum_{m \geq 1} S_{2m} = \frac{L^2}{4} - \frac{\pi^2}{48}, \tag{6.8}$$

$$\sum_{m \geq 1} 2^{2m-1} S_{2m} = L - 1, \tag{6.9}$$

$$\sum_{m \geq 1} 3^{2m-1} (S_{2m} + 3^{-2m}) = L - \frac{L^2}{4} + \frac{\pi^2}{48} - \frac{7}{12}. \tag{6.10}$$

Proof. The first two are $\mathcal{O}(1)$ and $\mathcal{O}(2)$, inside the Taylor disk of radius three. The $n=1$ term of the defining harmonic sum contributes $t/(t^2-9)$ to $\mathcal{O}(t)$. Subtracting it leaves a function analytic in $|t| < 5$. Its value at $t=3$ is

$$\text{FP}_{t=3} \mathcal{O}(t) - \frac{1}{12} = -P + L - \frac{1}{2} - \frac{1}{12},$$

which is the last right side. This also explains why adding 3^{-2m} before summation is necessary at this boundary point. $\square$

More generally, for an integer $K \geq 0$, set

$$T_p^{[K]} = S_p - \sum_{n=1}^K \frac{(-1)^n H_n}{(2n+1)^p}, \quad \mathcal{O}^{[K]}(t) = \mathcal{O}(t) - \sum_{n=1}^K \frac{(-1)^n H_n t}{(2n+1)^2 - t^2}.$$

Then $\mathcal{O}^{[K]}(t) = \sum_{m \geq 1} T_{2m}^{[K]} t^{2m-1}$ has Taylor radius exactly $2K+3$. Theorem 6.3 supplies its value at $t = 2K+2$ and its regular value at $t = 2K+1$ after the indicated finite rational pole contributions are removed. This gives an infinite family of convergent identities with elementary right sides. For $d = 2K+3$, $|t| < d$, and $M \geq 0$, a useful explicit remainder bound is

$$\left| \sum_{m > M} T_{2m}^{[K]} t^{2m-1} \right| \leq \frac{H_{K+1} |t|^{2M+1}}{d^{2M+2} (1 - |t|^2/d^2)}. \quad (6.11)$$

Indeed $H_n/(2n+1)^p$ strictly decreases for $n \geq 1, p \geq 2$, so the alternating-series estimate gives $|T_p^{[K]}| \leq H_{K+1}/d^p$. Summing the resulting geometric series proves the bound. The actual nonzero poles at $\pm d$ prove the exact Taylor radius.

These formulas impose exact analytic consistency constraints on an entire proposed sequence of fixed-weight reductions. They do not prove any individual S_{2m} reduction, because the resummations combine different weights. The distinction matters when they are used to audit conjectural coefficient families.

7 A new frozen mixed Gaussian candidate at weight fifteen

Put $g_{a,b} = \text{Im Li}_{a,b}(i, 1)$. The inspected incoming reports reach S_{12} ; the present search advances this particular candidate family to S_{14} . The old S_6 and S_8 relations, and the incoming S_{10} and S_{12} relations, remain conjectural. The coefficient table below completely specifies the new identity without rounding or an implicitly chosen numerical basis.

Conjecture 7.1 (Mixed Gaussian reduction at weight fifteen). *Let (X_j, c_j) be the ordered rows in Table 1. Then*

$$\sum_{j=0}^{15} c_j X_j = 0, \quad \text{equivalently} \quad S_{14} = - \sum_{j=1}^{15} \frac{c_j}{c_0} X_j. \quad (7.1)$$

For orientation, the first right-side coefficient and the last are

$$-\frac{c_1}{c_0} = \frac{28551728212443286}{20976315815914861}, \quad -\frac{c_{15}}{c_0} = -2.$$

The intermediate coefficients are part of the displayed vector and should not be replaced by decimals. Its entries have greatest common divisor one. The coefficient of $\beta(14)L$ matches the value -2 in the preceding frozen even-index candidates; this pattern is evidence for a possible uniform formula, not a proof of one.

7.1 Discovery and independent numerical audit

The exploratory evaluation used 3600 finite Euler weights computed as exact integers, with 1100 decimal digits in the subsequent arithmetic. PSLQ then ran at 850 digits, tolerance 10^{-800} , coefficient bound 10^{65} , and an iteration cap of 200000. It returned the primitive vector in Table 1. After that vector was frozen, its normalized residual at the retained 1100-digit precision was approximately 1.6223×10^{-1084} . This is a numerical diagnostic; Euler truncation and finite-precision arithmetic are not thereby made exact. The independent exact certificate below is the authoritative proximity statement.

Period X_j	Integer coefficient c_j
S_{14}	17137739346472321847995257166233600
$g_{14,1}$	-23326883533329005530721963055513600
$g_{12,3}$	-1241749496845608511840387596288000
$g_{10,5}$	577580508445935074901301710028800
$g_{8,7}$	-584626148672522170960472702976000
$g_{6,9}$	1032524260038200489807505575116800
$g_{4,11}$	-2814967862322082111057500831744000
$g_{2,13}$	16994562809106694872629900633702400
π^{15}	-969820884575582157963236591
$\beta(2)\zeta(13)$	2074025115468857154352052304000
$\beta(4)\zeta(11)$	23496914452549436838383360716800
$\beta(6)\zeta(9)$	112557931663509122499903946752000
$\beta(8)\zeta(7)$	319433131232914030174904411750400
$\beta(10)\zeta(5)$	765318186561145170729756524544000
$\beta(12)\zeta(3)$	3477679796290052414651630616576000
$\beta(14)L$	34275478692944643695990514332467200

Table 1: The complete primitive integer vector for the conjecture (7.1); the first row is $j = 0$.

A second calculation, at 110 decimal digits, evaluated the periods from Mellin integrals rather than the finite Euler sums. It used

$$S_{14} = -\frac{1}{13!} \int_0^\infty \frac{t^{13} e^{-t} \log(1 + e^{-2t})}{1 + e^{-2t}} dt$$

and, for $a \geq 2$,

$$g_{a,b} = \frac{1}{(a-1)!} \int_0^\infty t^{a-1} \operatorname{Im} \frac{ie^{-t} \operatorname{Li}_b(ie^{-t})}{1 - ie^{-t}} dt.$$

At $b = 1$ it used the integrated logarithmic formula

$$g_{a,1} = -\frac{1}{2(a-2)!} \int_0^\infty t^{a-2} \log(1 + e^{-2t}) \arctan(e^{-t}) dt.$$

The normalized residual was approximately -1.2378×10^{-110} . No rigorous quadrature-error enclosure is claimed for this numerical audit. It checks conventions and gives independent corroboration of the finite-sum calculation.

Proposition 7.2 (Exact rational proximity at weight fifteen). *For the actual periods in Table 1, put $R_{14} = \sum_{j=0}^{15} c_j X_j / c_0$. Then*

$$|R_{14}| < 10^{-775}. \quad (7.2)$$

The delivered rational interval contains zero. This proves proximity; the assertion $R_{14} = 0$ remains the conjecture.

7.2 Every analytic premise of the rational certificate

For a sequence f_n , define

$$\Delta^j f_0 = \sum_{n=0}^j (-1)^n \binom{j}{n} f_n, \quad E_N(f) = \sum_{j=0}^{N-1} 2^{-j-1} \Delta^j f_0.$$

A finite interchange of sums and Pascal's identity give

$$E_N(f) = 2^{-N} \sum_{n=0}^{N-1} (-1)^n f_n \sum_{r=n+1}^N \binom{N}{r}. \quad (7.3)$$

The positive weights inside the last sum are exact integers. This avoids an unstable floating-point finite-difference triangle.

For integer $a \geq 2, b \geq 1$, the ordinary Mellin formulas give

$$\frac{H_{2n}^{(b)}}{(2n+1)^a} = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^1 \int_0^1 \frac{(-\log x)^{a-1} (-\log y)^{b-1}}{1-y} \{(x^2)^n - (x^2 y^2)^n\} dy dx.$$

Keep the two terms grouped, especially at $b = 1$. Finite differences replace them by $(1-x^2)^j - (1-x^2 y^2)^j$. Summing the omitted Euler tail shows that $g_{a,b} - E_N(f)$ is the same positive Mellin measure applied to

$$2^{-N} \{W_N(x^2) - W_N(x^2 y^2)\}, \quad W_N(u) = \frac{(1-u)^N}{1+u}.$$

On $[0, 1]$, W_N decreases and $|W'_N| \leq N + 1$. Consequently

$$E_N(f) - \frac{N+1}{2^N 3^a} (1+2^{-b}) \leq g_{a,b} \leq E_N(f). \quad (7.4)$$

Indeed the difference of W_N values is between $-(N+1)x^2(1-y^2)$ and zero, and its integral is $-(N+1)3^{-a}(1+2^{-b})$. For $f_n = H_n/(2n+1)^p$, replace y^2 by y , obtaining

$$E_N(f) - \frac{N+1}{2^N 3^p} \leq S_p \leq E_N(f), \quad p \geq 2. \quad (7.5)$$

All interchanges follow either from absolute convergence of the original series at these integer orders or from the displayed integrable bound on the grouped Euler kernels.

The single-value sequences $(2n+1)^{-s}$ and $(n+1)^{-s}$ are moments of positive measures of mass one. Their omitted Euler tails are nonnegative and at most 2^{-N} . This encloses $\beta(s)$ and $\eta(s)$; multiplying by $(1-2^{1-s})^{-1}$ encloses $\zeta(s)$ when $s > 1$. Finally, use

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239), \quad L = 2 \sum_{n \geq 0} \frac{1}{(2n+1)3^{2n+1}}.$$

The first omitted term encloses each alternating arctangent series. For M terms of the logarithm series, its positive tail is at most $9/[4(2M+1)3^{2M+1}]$. Machin's identity uses the real arctangent: the tangent of $4 \arctan(1/5) - \arctan(1/239)$ is one, and this angle lies in $(0, \pi/2)$, fixing its value as $\pi/4$.

Proof of Proposition 7.2. The standalone program `code/verify_s14.py` reconstructs all sixteen period intervals using $N = 2600$, 650 terms in each arctangent series, and $M = 950$ logarithm terms. For the nested sums it takes the common denominator $D = \text{lcm}(1, \dots, 2N)$: each harmonic numerator and each power denominator in (7.3) can then be accumulated as an integer before one final rational division. No decimal input, numerical period evaluation, PSLQ calculation or assumed identity enters the replay.

The verifier reads the frozen integer vector, forms the normalized interval dot product by exact rational interval arithmetic, and compares its endpoints to $\pm 10^{-775}$. These comparisons are strict, and the interval contains zero. It also compares every reconstructed basket interval and both final endpoints against `data/s14_rational_certificate.json`. This is a finite proof of the enclosure (7.2) using only the analytic bounds just established. It supplies no mechanism for replacing a zero-containing interval by exact equality. $\square$

7.3 Provenance, replay and the remaining proof problem

The common finite-Euler arithmetic is a small adaptation of the incoming uniform-continuation report's exact proximity implementation, and its error theorem is retained rather than claimed as new. The inspected archive is `proveit_polylogarithms_research_2026-10-10 (1).zip` in the pinned repository. Source-member and archive SHA-256 digests are recorded in `data/mixed_gaussian_provenance.json`. The new deliverables are the S_{14} vector, its exact proximity certificate, its independent quadrature audit, and the rational sampling/finite-part theorems above.

From the package root, the exact certificate is replayed by

```
python code/verify_s14.py
```

It requires only the Python standard library. Optional diagnostics use `code/audit_s14.py`, `code/search_s14.py`, and `code/verify_mixed_generator.py`; these require `mpmath`. The search is separate from the proof of proximity.

A useful next target is a uniform functional identity that implies the fixed-weight conjectures simultaneously. The existing restricted single-product obstruction remains valid: it excludes only that stated row span. The known weight-five octahedral certificate supplies information outside it. The higher-weight problem is therefore to give an equally explicit new relation or an independently replayable finite certificate, not to infer global irreducibility from the restricted obstruction. The exact resummations above offer consistency constraints for any proposed uniform formula, while the S_{14} vector supplies a fresh high-weight target.

8 Audit findings, status changes, and reproducibility

8.1 What has changed mathematically

The source snapshot matters because several incoming reports overlap, and some questions in an earlier report are answered by a later one. We inspected the canonical manuscript and the textual contents of all six incoming archives at the pinned revision. We used that comparison to choose the present targets and to identify the inherited inputs. This is an audit of the dependencies and claims relevant to this continuation, not a new line-by-line certification of all 379 pages.

The main changes of status are as follows.

Question in the snapshot	Previous status	Result in this article
Global limit of the optimal kernel constants	Local scaling theorem; global limit conjectured in [2, Section 8]	Theorem 2.3 proves the limit, the limiting maximizer, and a uniform quantitative enclosure.
Uniqueness and decrease of kernel maxima	Conjectured at every finite truncation	Theorem 2.8 proves uniqueness at $N = 2$; Theorem 2.5 proves eventual uniqueness, decrease, and a convergent expansion.
Sharp decay of $C_N - 1$	A lower asymptotic; matching upper asymptotic conjectured in [3, Section 9]	Theorems 3.1 and 3.3 prove the global asymptotic and the eventual axis reduction.
Next order scale and near maximizers	Not resolved by the leading candidate	Theorem 3.5 gives the threshold-layer correction; Corollary 3.9 gives the exact $o(N^{-p})$ criterion.
Polynomial Appell root separation	Exponential lower bound; a polynomial bound requested in [4, Section 8]	Theorem 4.5 proves $\text{mesh}(Q_n) \geq 1/(10n^2)$.
Growing-index Lerch saturation	Explicit exponential threshold and logarithmic index range	Theorem 4.1 gives an $O(n^4 \log^2 n)$ threshold and a uniform $8/\sqrt{k} \log$ error.
Signs at outer orders $-2, -3$	Individual opposite-sign evaluations; negativity proved down to -1	Theorem 5.3 classifies all $b > 0$: one simple crossing in each case.
Mixed Gaussian periods	S_4 proved; S_6, S_8, S_{10}, S_{12} conjectural	Exact generating and sampling identities are proved. A new S_{14} candidate is supplied with certified proximity; its equality remains conjectural.

The constants M_N and C_N must remain distinct. The former is the maximum of a pointwise two-variable kernel; the latter is the supremum after averaging that kernel against the particular gamma distributions coming from the two polylogarithm orders. Although $C_N \leq M_N$, their limits are different:

$$M_N \longrightarrow 1.05386193036\dots, \quad C_N \longrightarrow 1.$$

The limiting pointwise extremizer has $Np_N \rightarrow c_\infty$ and $y_N \rightarrow y_\infty$. The order-averaged extremizer instead approaches the outer-order boundary, with $b_N \sim \log N / \log(3/2)$. The probability laws allowed by the order parameters do not reproduce an arbitrary point mass at the pointwise maximizing pair.

8.2 Proposed corrections and precise integration instructions

No theorem-level error was established in the inherited results used in these proofs. The necessary manuscript changes are chiefly updates of status and explicit qualifications. They should not be presented as retractions of correct earlier results.

The two Euler conjectures. In [2], replace the conjectural status of `conj:limit` by Theorem 2.3. Retain the all-finite-index content of `conj:maxima`: the present eventual theorem and the exact $N = 2$ case leave intermediate finite indices unclassified. In [3], replace `conj:euler-rate` by Theorem 3.3, and add the exact axis-reduction theorem before discussing the rate. It supplies the global upper bound that an axis computation alone cannot give.

The strict domain-specific estimate $R_N(a, b) < 1$ for $a \geq 1$ in the canonical `chapters/05-real-euler.tex` is an explicit dependency. The larger universal constant C_* does not supply the same exclusion. The already proved value $C_* = 1.1365611033395 \dots$ remains a separate, all-index optimum.

Quantifiers on asymptotics. The uniqueness and monotonicity statements for the two maximizing problems are eventual unless explicitly stated at $N = 2$. We do not provide a numerical starting index for those eventual statements. Also, the next Taylor moment of the axis kernel is not the next asymptotic term of the optimized remainder. The threshold layer gives $N^{-\kappa}(\log N)^{-1/2}$, and the first relative inverse-log coefficient is large in magnitude. Remark 3.6 explains why moderate-index agreement is an unreliable way to assess this second term. This is a qualification needed in further extrapolation, not a claim that the incoming reports made the incorrect expansion.

Lerch geometry. In [4, Section 4], retain the original mesh bound: it is useful at small degrees and is compatible with the stronger polynomial estimate. Replace an exponential threshold as the only available effective theorem by Theorems 4.1 and 4.9. The sharper generic transfer can use any proved mesh bound, including the maximum of the old and new lower bounds. The family at $\rho = 1$ is the derivative family with $k \geq 1$; the undifferentiated Lerch sum need not converge there. The polynomial growing-index theorem establishes leading relative locations and the full zero count, not uniform constant offsets or complete zero expansions in the joint limit.

Negative outer orders. Keep the previously proved negativity for every real outer order at least -1 . The two formulas at $b = 1$ used in Theorem 5.3 were already in the incoming Euler report and should retain that provenance. What is new is the unique simple transition throughout $b > 0$. The kernel q_5 shows why the one-crossing proof cannot simply be asserted for all lower orders.

Arithmetic status and old labels. The canonical `chapters/04-S4-proof.tex` labels `s4proof:thm:s4` and `gaussian:thm:S4` denote an already proved result; the older `gaussian:conj:S4` survives as a compatibility label, not as evidence that the identity is still open. By contrast, `cycloquot:conj:S6` in `chapters/04-cyclotomic-quotients.tex` and `s8new:conj:S8` in `chapters/04-S8-candidate.tex` remain conjectures. The S_{10}, S_{12} vectors in [4, Section 5] are prior candidates. Our weight-fifteen vector is new relative to those files; restricted relation-space calculations at the same weight were already present.

A rational enclosure containing zero proves a bound on the residual, not its exact vanishing. Likewise, nonmembership in a specified finite relation space does not prove independence of complex periods or impossibility of a reduction using a larger alphabet. The generating

identities of Section 6 also do not authorize coefficientwise fixed-weight conclusions beyond what their proofs actually establish.

The companion integration note gives the source files and labels in machine-searchable form. Labels in this article use the prefixes `ker:`, `ord:`, `mesh:`, `neg:`, and `mix:` to reduce collision risk when sections are moved.

8.3 What the computational artifacts prove

The package separates exact replay from numerical diagnostics. The distinction is mathematical: increasing working precision is not a substitute for bounding an infinite tail, and a computed optimizer is not a global maximum certificate.

1. **Exact rational kernel enclosures.** `code/verify_kernel.py` isolates the root defining y_∞ using rational bounds for the logarithm, encloses the corresponding exponential constant, and isolates the algebraic $N = 2$ root and its rationally defined maximum. The qualitative uniqueness and global conclusions are supplied by Section 2. The separate diagnostic script checks symbolic elimination and plots locally computed stationary values.
2. **Exact exponent comparisons.** `code/verify_order_constants.py` uses rational atanh-series bounds for logarithms to check the strict comparisons between the exponents in Section 3. These finite sign checks support the stated separation of scales; the uniform analytic arguments remain part of the proof.
3. **Appell and transfer arithmetic.** `code/verify_mesh.py` records exact algebraic and rational checks used in the mesh constants and computes the sufficient integer thresholds without rounding them downward. Its finite examples are identified as checks of formulas, not a replacement for the argument valid at every degree.
4. **The frozen weight-fifteen certificate.** `code/verify_s14.py` starts from the integer vector and uses Python integers and rational fractions throughout. Its finite Euler sums, elementary constant intervals, and analytic tails produce the shipped interval for R_{14} and verify $|R_{14}| < 10^{-775}$. Both endpoints and all basket intervals are replayed exactly. The certificate contains zero and does not prove $R_{14} = 0$.
5. **Independent diagnostics.** Symbolic differentiation verifies the negative-order kernels, and separate Dirichlet and Mellin evaluations check their displayed decimals. The mixed-generator program checks rational samples, integer samples, pole finite parts, and resummation tails. A separate Mellin calculation checks the frozen S_{14} vector without using its discovery summation. These high-precision computations are corroboration, with their precision and residuals recorded in the data files.

The default replay uses only the Python standard library for its exact certificates. Optional numerical and symbolic diagnostics use `mpmath`, `sympy`, and `matplotlib`; the versions used in this delivery are recorded. The main `article.tex` builds with pdfLaTeX and standard mathematics packages. The source revision, incoming archive hashes, relevant source hashes, verification commands, and PDF review are included in the package. No repository modification or publication is part of this delivery.

9 Further research questions

The results above remove several qualitative obstacles, but they also identify where the present methods lose information. The following questions are ordered by their relation to the proved theorems and by the kind of additional argument they require.

9.1 Finite-index extremizers and the order boundary

1. Is every K_N uniquely maximized, and is M_N strictly decreasing for every $N \geq 2$? The exact $N = 2$ elimination and the analytic large- N theorem leave a finite but presently unbounded intermediate region. One route is to obtain a quantitative neighborhood and starting index for the implicit-function argument, then certify the remaining rational kernels by elimination or interval subdivision. A more satisfying answer would be a direct comparison principle in N . Pointwise monotonicity of a fixed-order normalized remainder is unavailable in general, and monotonicity of the auxiliary envelope does not alone compare its underlying maxima.

2. Does exact axis reduction hold at every truncation? For sufficiently large N , every positive outer order is strictly inferior to the axis. The full finite-index question is to decide whether $C_N = \max_b A_N(b)$ holds for all N , and whether the axis maximum is always unique. If it fails, identify the first interior maximizer and its transition. The uniform comparison (3.7) is a useful starting inequality, but its positive and negative terms are too coarse at modest indices to decide the question without further work.

3. Make the eventual statements effective. Produce explicit starting indices for axis reduction, axis uniqueness, kernel uniqueness, and the respective monotonicity results. The present proofs give concrete inequalities but leave several *sufficiently large* transitions unevaluated. Quantitative gamma tail estimates and explicit bounds on derivatives of the limiting kernels could turn them into finite gates. The practical challenge is to obtain useful indices rather than enormous formal upper bounds.

4. Describe the threshold regimes beyond the maximizing order. The Mellin integral in Lemma 3.4 converges for $3/2 < \alpha < 5/2$. At the endpoints, a Taylor subtraction collides with the integrability boundary. Determine uniform formulas as $b/\log N$ approaches these values at a rate depending on N , and determine the subsequent subtraction hierarchy. The poles of the continued gamma factor suggest where logarithmic transition terms should enter, but the integral must be treated before any such expansion is asserted.

5. Obtain usable remainder bounds at the second scale. The inverse-log coefficient near the maximizing slope is large, so a few formal terms are not a reliable moderate- N approximation. Find an expression retaining the threshold integral, or another uniform approximation, with a computable relative error. This would connect the exact extremizer theorem with practical Euler truncation budgets. It would also allow meaningful numerical tests of the optimizer displacement well before the extremely late asymptotic regime.

9.2 Appell spacing and joint zero geometry

6. Determine the true minimum-gap asymptotic. The bound $1/(10n^2)$ answers the polynomial-separation question, but it is not expected to describe every feature of the root configuration. Determine the actual order and leading constant of $\text{mesh}(Q_n)$, and locate the pair of roots where the minimum occurs. Bulk zero distributions alone need not control the smallest spacing, especially if it occurs at an edge. A result resolving this distinction would feed directly into the scale-free transfer theorem.

7. Improve the remote-block method. The proof uses $3n$ factors near $15n^2$ because that choice allows elementary Laguerre estimates with transparent constants. Optimize the block length, its position, and the ordered-root comparison. Can information on correlated

root motion reduce the loss from the worst displacement of one endpoint? Can several blocks yield a stronger power of n ? The mesh-preservation theorem should remain separate from any unsupported monotonicity in an individual factor coefficient.

8. Find the optimal saturation scale and first offsets when n grows. The present sufficient threshold is $O(n^4 \log^2 n)$, uniformly over ρ and every root. It is not a lower bound on the first derivative order where saturation actually occurs. Determine the true scale, its dependence on the branch, and whether the harmonic factor is essential. Then strengthen the leading relative location to a uniform additive offset and a controlled higher expansion. This requires bounds on root-conditioned derivatives, not merely separation and total zero count.

9. Extend the transfer theorem to other positive weights. Its essential input is a bound on the logarithmic derivative of the positive weight, together with a real-rooted factor in $\log t$. Identify other polylogarithmic and zeta derivative families with that structure, including deformations whose logarithmic Lipschitz constant grows with the degree. Determine when the resulting constants still permit a polynomial joint-growth range. Sign-changing weights require additional control because the global variation bound may then change.

9.3 Sign transitions and exact period reductions

10. Classify every negative integer outer order. The finite product formula gives an explicit rational Gaussian kernel for every m . Determine its number of roots in $(0, 1)$, then determine which permitted sign changes survive its gamma average as b varies. The $m = 5$ example already separates the one-crossing mechanism from the general problem. Possible tools include reciprocal-polynomial substitutions, Eulerian polynomial recurrences, and Mellin variation arguments. A root count for the rational kernel is only an upper bound for the number of transitions in the averaged function.

11. Construct a generating identity for the fixed-weight mixed Gaussian reductions. The exact identities for $\mathcal{S}$ and $\mathcal{O}$ evaluate infinite sums across weights. They do not supply the finite weight-by-weight relation needed for S_6 or S_{14} . The central problem is to find a bivariate generating relation, a differential equation with enough boundary data, or an explicit iterated-integral reduction whose coefficients are the desired finite identities. Rational sampling may constrain such a construction and indicate which cyclotomic levels it naturally uses.

12. Prove or refute the frozen S_{14} relation. The interval in Proposition 7.2 is a precise target for symbolic work. A proof should exhibit an exact algebraic or iterated-integral relation, including endpoint terms and branch conventions. Further digits would only strengthen proximity. If the current finite relation space is insufficient, specify the enlarged alphabet or class of relations and test the frozen vector there. Do not infer period independence from a restricted rank calculation.

13. Determine the arithmetic structure of the coefficient vectors. The previously recorded S_{10}, S_{12} candidates and the present S_{14} vector have a common basket pattern. Establish whether there is a uniform rational formula, recurrence, or generating function for these coefficients, and explain the persistent $-2\beta(2m) \log 2$ term where it occurs. Searches should freeze their basket and vector before independent evaluation and should report exact residual intervals with an explicit normalization. Any conjectured coefficient law needs testing outside the weights used to infer it.

9.4 Reproducible and formal continuations

14. Formalize small reusable proof components. Several arguments here are natural proof-assistant targets: the divided-difference expectation, the elementary global maximum criterion, the rational $N = 2$ elimination certificate, the ordered-root comparison for $1 + c\partial$, and the gamma log-moment bound. They can be separated from the heavier complex analytic continuation machinery. A formal integration plan should distinguish algebraic certificate checking from formalized analysis and should retain the exact source revision for inherited dependencies.

The first mathematical priorities are an effective finite-index extremizer classification, a sharper Appell mesh theorem, and an exact relation producing the fixed-weight mixed Gaussian identities. Each has a concrete obstruction exposed by the present proofs, and none is resolved by additional numerical precision alone.

References

- [1] V. Reshetnikov and ProveIt contributors, *Polylogarithms manuscript*, in the ProveIt repository, revision `4c173c06cc32c9cea554b39be1837ad2ae897fc1`, 10 October 2026. [Immutable manuscript snapshot](#). The accompanying provenance manifest records the exact files used.
- [2] ProveIt research continuation, *The Sharp Universal Euler Constant for Harmonic Polylogarithms: Positive Divided Differences, Exact Certificates, and All-Order Error-Moment Identities*, 10 October 2026. Archive `ProveIt_Sharp_Universal_Euler_2026-10-10.zip`, in `docs/incoming` at the revision in [1]. Principal dependencies: Sections 2–3, 6 and 8.
- [3] ProveIt research continuation, *Polylogarithms: Uniform Bounds and Structural Transitions*, 10 October 2026. Archive `ProveIt_Polylogarithms_Research_2026-10-10.zip`, internal directory `polylogarithms_uniform_bounds_20261010`, in `docs/incoming` at the revision in [1]. Principal dependencies: Section 2 and the conjecture `conj:euler-rate` in Section 9.
- [4] ProveIt research continuation, *Sharp Euler Bounds and Uniform Asymptotics in Polylogarithmic Analysis: Reflected Moments, Lerch Zeros, and Integral Shuffle Lattices*, 10 October 2026. Archive `proveit_polylogarithms_research_2026-10-10(1).zip`, internal directory `polylogarithms_uniform_continuation`, in `docs/incoming` at the revision in [1]. Principal dependencies: Sections 4–5 and 8.
- [5] ProveIt research continuation, *Sharp Euler Bounds, Golden Ladders, and Double Radial Turning Points*, 10 October 2026. Archive `proveit-polylogarithms-research-2026-10-10.zip`, internal directory `proveit-polylogarithms-research-2026-10-10`, in `docs/incoming` at the revision in [1]. Its golden identities and constructive Cayley calculations are prior work, not claims of the present article.
- [6] ProveIt research continuation, *A Unique Quartic Transition, Möbius Rigidity, and Two Radial Turning Points*, 10 October 2026. Archive `ProveIt_Polylogarithms_Radial_Bifurcation_2026-10-10.zip`, in `docs/incoming` at the revision in [1].
- [7] ProveIt research continuation, *Signed-Kernel Oscillation at Every Depth: Minimal Pick Compensation, Zero Geometry, and Certified Polylogarithmic Identities*, 10 October 2026. Archive `ProveIt_AllDepth_Oscillation.zip`, in `docs/incoming` at the revision in [1].

- [8] NIST Digital Library of Mathematical Functions, Sections 3.9 (Euler transformation), 5.8 (reciprocal-gamma products), 5.9 (gamma integrals), 5.11 (Stirling expansions), 18.5, 18.8 and 18.9 (Laguerre representations, differential equations and recurrences), and 25.12 (polylogarithms). Accessed 10 October 2026. <https://dlmf.nist.gov/>.
- [9] P. Brändén, I. Krasikov, and B. Shapiro, *Elements of Pólya–Schur theory in the finite difference setting*, Proceedings of the American Mathematical Society **144** (2016), 4831–4843. doi:10.1090/proc/13115. Theorem 1 records the classical Riesz mesh-preservation theorem. Author-hosted full text: <https://staff.math.su.se/shapiro/Articles/FiniteDifference.pdf>.